VOILA: Value-of-Information Guided Fidelity Selection for Cost-Aware Multimodal Question Answering

Anonymous ACL submission

Abstract

Multimodal retrieval-augmented systems store images at multiple fidelities—captions, thumbnails, compressed JPEGs, full-resolution originals—yet retrieve at fixed resolution regardless of the query, paying maximum acquisition cost even when a coarser representation would suffice. We formalize *pre-retrieval fidelity selection*: given only the question text, select the cheapest fidelity whose expected accuracy justifies its retrieval cost, *before any image is accessed*. This problem is distinct from post-retrieval cascades (FrugalGPT, RouteLLM) that act after context is fetched, and from concurrent resolution-selection methods that still require image access. We show the problem is tractable from question text alone: linguistic structure (e.g., *how many*, *what color* vs. *is there*, *does it show*) strongly predicts required visual fidelity. VOILA estimates per-fidelity success probabilities via a calibrated gradient-boosted classifier over question features, then applies a greedy value-of-information rule—with no image access and no VLM execution at decision time. Across five datasets and six VLMs (7B–235B parameters), VOILA reduces retrieval cost by 50–65%, matching or exceeding full-resolution on OCR-heavy and domain-specialized benchmarks while exposing a controllable accuracy–cost trade-off on general VQA benchmarks.

1 Introduction

Consider two questions posed about the same image: *"Is there a car in the photo?"* and *"What does the license plate say?"* The first can be answered from a heavily compressed thumbnail; the second requires reading fine-grained text at full resolution. This distinction—*what visual fidelity a question demands*—is not random: questions containing words like *how many*, *what color*, *read the text*, or *which airline* signal subtasks (counting, color discrimination, OCR, logo recognition) that require high-resolution evidence, while questions containing *is there*, *does the image show*, or *what type* are typically resolvable from low-fidelity representations. We show that this linguistic predictability is strong enough to drive a principled pre-retrieval fidelity selection system, enabling significant cost reductions with a controllable accuracy–cost trade-off.

Despite this predictability, most VLMs retrieve images at a fixed resolution regardless of the query (Alayrac et al., 2022; Li et al., 2023; Liu et al., 2023; Bai et al., 2023). Model routing methods (Chen et al., 2023; Ding et al., 2024; Ong et al., 2024) reduce inference cost but make their decision *after* context is retrieved, assuming all fidelities are freely accessible. In practice, images are stored across heterogeneous tiers at different retrieval costs (Lonn et al., 2019; Amazon Web Services, 2024b); retrieval cost is incurred *before* any model runs, so post-hoc routing cannot avoid it. Concurrent resolution-selection work (Kimhi et al., 2025) still requires running a compact VLM on the image; VOILA routes from question text *before any image is accessed*.

We formalize this as *pre-retrieval fidelity selection* and introduce **VOILA** (Value-Of-Information guided Latency-Aware fidelity selection): a gradient-boosted classifier over question features estimates per-fidelity correctness probabilities, isotonic calibration converts raw scores into reliable probabilities, and a greedy value-of-information rule selects the minimum-cost sufficient fidelity—all before any retrieval cost is incurred, with regret bounded by calibration error (Appendix C).

This paper makes the following contributions:

- **Problem formulation:** Pre-retrieval fidelity selection for multimodal RAG—distinct from post-retrieval cascades and from concurrent methods that still require image access—with a near-Bayes-optimal greedy policy under calibrated estimation (§3, Appendix C).

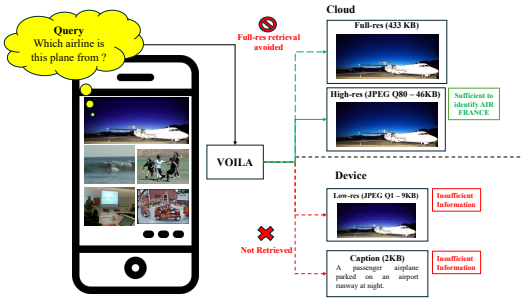


Figure 1: VOILA predicts the minimum visual fidelity required to answer a query *before* any image is retrieved. For "Which airline is this plane from?"—an OCR-dependent question—VOILA selects a mid-fidelity JPEG (46 KB) that preserves the airline logo, avoiding full-resolution retrieval (433 KB) and achieving a 9× bandwidth reduction.

- **Empirical finding:** Question linguistic structure (*how many*, *what color* vs. *is there*, *does it show*) is a strong, sufficient signal for fidelity routing without image access (§2).
- **VOILA framework:** A model-agnostic GBR + isotonic-calibration system running in 0.45 ms per query on CPU, with no VLM involvement at decision time (§3).
- **Evaluation** across six VLMs (7B–235B) and five datasets: 50–65% cost reduction, matching or exceeding full-resolution on OCR-heavy and specialized benchmarks, within 10 accuracy points (average) on general VQA.

2 Question Semantics Predict Visual Information Requirements

We present three empirical observations that collectively establish (i) the practical importance of the pre-retrieval fidelity selection problem and (ii) the tractability of solving it from question text alone. All observations are based on VQA-v2 with Pixtral-12B across input fidelities ranging from text-only captions to full-resolution images (Figure 2), with qualitatively identical patterns across all six evaluated models.

Observation 1: Fidelity Requirements Are Linguistically Predictable. Figure 2a shows that query type strongly predicts minimum required fidelity. Questions containing *how many*, *what color*, or *read/what does it say* signal subtasks requiring fine-grained evidence (object counting, hue discrimination, OCR) that low-fidelity representations destroy; questions containing *is there*, *does the image show*, or *what type* involve coarse semantic judgments resolvable from captions or thumbnails. On VQA-v2, yes/no and color questions plateau at JPEG Q10 (cost 53.2), while counting, OCR, and spatial queries require full resolution (cost 100). No single fidelity dominates, establishing that fidelity selection is both necessary and tractable from question text alone (Kafle and Kanan, 2017; Agrawal et al., 2018). Under-escalation errors (23–25% of queries, §5) reflect cases where the correct fidelity depends on visual context rather than linguistic structure—the main reducible component of the Oracle–VOILA gap.

Observation 2: Post-Retrieval Confidence Is an Unreliable Escalation Signal. FrugalGPT-style cascades (Chen et al., 2023) retrieve low-cost context first and escalate when VLM log-probability falls below a threshold. This fails in the fidelity domain: as Figure 2b shows, incorrect low-fidelity answers receive log-probabilities comparable to—or exceeding—correct ones (43% of incorrect caption-fidelity answers on VQA-v2 exceed the median log-probability of correct caption-fidelity answers). For Qwen2-VL-72B, the log-probability AUC for caption-to-full escalation is 0.338—*below random* (ROC curves in Appendix F)—confirming VLM confidence is an actively misleading escalation signal at low fidelity.

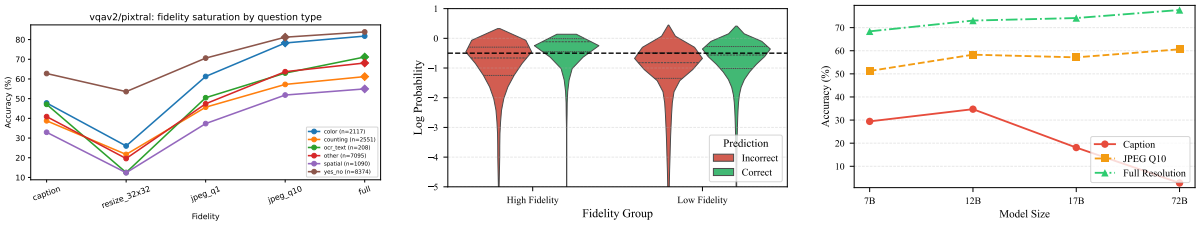
Observation 3: Model Scaling Does Not Substitute for Sufficient Visual Information. Figure 2c shows that fixing input fidelity at a low level caps accuracy regardless of model size: caption-only accuracy saturates or *degrades* at larger scale (likely due to hallucination when visual grounding is absent), while full-resolution accuracy improves consistently with scale. Model scaling and information acquisition are therefore *complementary rather than substitutable* optimization axes—the key motivation for pre-retrieval fidelity selection.

These observations establish that fidelity requirements are (1) linguistically predictable pre-retrieval, (2) not estimable from post-hoc confidence, and (3) not compensated by model scaling—motivating VOILA's design.

3 VOILA: Query-Conditioned Fidelity Selection

3.1 Problem Formulation

Let (q, x) denote a visual question q paired with an image x. We assume access to a discrete, ordered set of input fidelities $\mathcal{F} = \{f_1, f_2, \ldots, f_K\}$, sorted by increasing acquisition cost, where each fidelity



(a) Each curve traces accuracy vs. fidelity for one question type; stars mark the first fidelity achieving 95% of full-resolution accuracy. Counting and OCR tasks require full resolution; yes/no and color saturate at JPEG Q10 (cost 53.2).

(b) VLM output log-probabilities are unreliable indicators of context sufficiency: incorrect low-fidelity answers receive confidence scores comparable to correct ones.

(c) Model scaling does not compensate for insufficient fidelity: accuracy saturates or degrades at low fidelity regardless of model size.

Figure 2: Three empirical observations on VQA-v2 with Pixtral-12B that motivate VOILA's design: (a) question type reliably predicts fidelity requirements; (b) post-hoc confidence signals are unreliable for escalation decisions; (c) model scaling cannot substitute for adequate visual information.

$f \in \mathcal{F}$ incurs a cost $c(f) > 0$ capturing retrieval latency, bandwidth, and compute overhead. Lower fidelities correspond to inexpensive representations (e.g., captions or thumbnails), while higher fidelities provide richer visual information (e.g., full-resolution images).

For fidelity f, if the vision–language model (VLM) produces an answer $\hat{a}_f$, our objective is to select the fidelity that maximizes expected utility:

$$f^\star = \arg\max_{f \in \mathcal{F}} \underbrace{\Pr(\hat{a}_f \text{ is correct} \mid q)}_{\text{expected accuracy}} - \underbrace{\lambda c(f)}_{\text{acquisition cost}},$$

where **the probability of correctness must be estimated *without* retrieving fidelity f or running the VLM at that fidelity**, and λ controls the accuracy-cost trade-off. VOILA accomplishes this through question-conditioned learned estimators.

3.2 Design Overview: A Three-Stage Pipeline

VOILA adopts a three-stage pipeline: (1) **question-conditioned success scoring** (§3.3) predicts, from question features alone, how likely each fidelity is to yield a correct answer; (2) **fidelity-specific calibration** (§3.4) converts raw scores into reliable probabilities; and (3) **VOI selection** (§3.5) picks the minimum-cost fidelity whose expected accuracy gain justifies retrieval cost.

3.3 Question-Conditioned Success Scoring

Fig. 2 illustrates that question features contain predictive signal about cross-fidelity performance *before* any visual information is retrieved. To exploit this signal, we construct training data by executing the VLM at each fidelity f on labeled examples and recording whether the produced answer is correct. Let $y_f \in \{0, 1\}$ denote the correctness label for fidelity f. We extract a question feature representation $\phi(q)$ consisting of:

- TF–IDF features of the question text (top-5,000 unigram vocabulary computed over the training split),
- lexical statistics: question length, presence of numeric tokens, superlatives, and negations,
- question-type indicators: five binary flags derived by rule-based matching on question-initial words and syntactic patterns (counting: *how many*; yes/no: *is/are/does/do*; color: *what color*; spatial: *where*; OCR/text: *read/spell/say/written*).

We deliberately use lightweight, interpretable features rather than large pretrained encoders (e.g., BERT or sentence transformers) for two reasons. First, it demonstrates that the task is solvable from surface-level linguistic structure alone—the central empirical claim of this paper. Second, TF-IDF feature extraction requires no GPU and completes in under 1 ms, making VOILA deployable on edge devices without a pretrained language model dependency. The combined feature vector has 5,017 dimensions (5,000 TF-IDF + 12 lexical + 5 question-type). Figure 5 (Appendix F) visualizes the top-ranked TF-IDF features: words predictive of caption sufficiency (*is*, *there*) differ systematically from those predicting full-resolution need (*color*, *many*, *sport*).

Feature ablation experiments (Table 3, §5) confirm that the full 5,017-dimensional combination achieves the best in-distribution accuracy–cost Pareto position. Crucially, question-type

flags alone are competitive under cross-dataset transfer: trained on VQA-v2 and evaluated on GQA, TextVQA, and FloodNet without retraining, they retain 79–92% of in-domain routing accuracy, whereas TF-IDF vocabulary degrades more sharply under domain shift because it encodes corpus-specific word statistics rather than language-universal syntactic roles (on FloodNet, question-type-only routing achieves 53.1% OOD accuracy vs. 42.7% for TF-IDF-only). The TF-IDF vocabulary size of 5,000 is a conservative high-capacity default; ablation experiments show no monotonic gain from increasing vocabulary size (performance is comparable at 1,000–2,000 tokens), so 5,000 is retained as a robust upper bound.

Using these features, we train a fidelity-specific regressor

$$r_f(q) = g_f(\phi(q))$$

to produce a raw score that correlates with the likelihood that the VLM will answer correctly at fidelity f. **Crucially, g_f is a learned estimator of VLM behavior, not the VLM itself.** The regressor learns statistical patterns between question characteristics and cross-fidelity success rates during offline training. At inference time, **computing $r_f(q)$ requires only a forward pass through a lightweight tree ensemble—no visual retrieval, no VLM execution, no multimodal processing**. This is orders of magnitude faster and cheaper than actually running the VLM at each fidelity.

We use a Gradient Boosting Regressor (Prettenhofer and Louppe, 2014) for g_f: it handles high-dimensional sparse TF-IDF inputs naturally and captures non-linear question–fidelity interactions (e.g., "counting + small objects" ⇒ high fidelity) without manual feature engineering. Predictor ablations (§5) confirm calibration quality matters more than model choice; GBR, logistic regression, ridge, and MLP all perform similarly under isotonic calibration.

3.4 Fidelity-Specific Probability Calibration

The regressor output $r_f(q)$ is *not* a calibrated probability; rather, it is a *question-conditioned success score* that orders queries by their relative likelihood of being answered correctly by the vision–language model at fidelity f. Because $r_f(q)$ is not probabilistic, we apply fidelity-specific post-hoc calibration using isotonic regression (Guo et al., 2017). For each fidelity f, we fit an isotonic regression model

$$\psi_f : r_f(q) \mapsto \hat{p}_f(q),$$

on a held-out 20% split of the training data (disjoint from both GBR training and evaluation test sets), where $\hat{p}_f(q) \in [0, 1]$ estimates the probability that the vision–language model would answer question q correctly if executed at fidelity f. All datasets are split 75%/25% into training and evaluation sets; within the training portion, 80% is used for GBR training and 20% for isotonic calibration (i.e., 60% / 15% / 25% of the full dataset). Isotonic regression is non-parametric and fits in $O(n \log n)$ time; at the dataset sizes we use (3,000–12,000 examples per fidelity), it does not require cross-validation and is not prone to overfitting. By fitting a non-decreasing mapping from raw scores to empirical success frequencies on held-out data, isotonic regression enforces monotonicity by construction and corrects systematic over- and under-confidence in the raw scores. This calibration step uses only $(r_f(q), y_f)$ pairs, where $y_f \in \{0, 1\}$ denotes historical correctness at fidelity f. Isotonic regression is non-parametric (no assumptions on miscalibration shape), preserves the score rank ordering learned by the regressor, and has well-understood convergence guarantees (Guo et al., 2017) (formal bounds in Appendix C.2). **Cross-fidelity monotonicity.** The per-fidelity calibrators above enforce monotonicity *within* each fidelity's score-to-probability mapping, but do not guarantee that $\hat{p}_{f_1}(q) \le \hat{p}_{f_2}(q) \le \cdots \le \hat{p}_{f_K}(q)$ holds *across* fidelity levels for a given query. In practice, average accuracy need not be monotone in fidelity (e.g., on TextVQA, VLM-generated captions often score higher than low-quality thumbnails because they already encode scene text). To guarantee that the greedy escalation policy of Algorithm 1 finds the global utility optimum, we apply a lightweight secondary isotonic regression pass *across* fidelity levels on the per-query calibrated scores at inference time, enforcing the ordering $\hat{p}_{f_1}(q) \le \cdots \le \hat{p}_{f_K}(q)$. This step adds negligible overhead (<0.01 ms) and removes non-unimodal utility cases that would otherwise cause the greedy break to terminate prematurely.

The key effect of isotonic calibration is probability reliability. Figure 3 shows reliability diagrams for VQA-v2/Pixtral-12B: isotonic calibration closely tracks the perfect-calibration diagonal, while raw GBR scores and VLM log-probabilities deviate substantially (ECE: $0.025 \to 0.012$; Brier: $0.205 \to 0.203$). When $\hat{p}_f(q)$ reliably estimates the true correctness probability, the VOI gain $\hat{p}_{f'}(q) - \hat{p}_f(q)$ becomes an accurate predictor of the

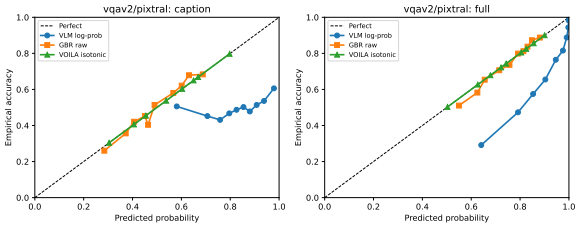


Figure 3: **Reliability diagram** (VQA-v2, Pixtral-12B). Isotonic calibration closely tracks the perfect-calibration diagonal (dashed); raw GBR scores and VLM log-probabilities deviate substantially, causing log-prob cascades to under-escalate (Observation 2).

accuracy improvement from escalation. Additional model–dataset pairs are in Appendix F (Figure 7).

3.5 Value-of-Information–Based Fidelity Selection

With calibrated success probabilities $\hat{p}_f(q)$ in hand—**computed from question features alone, without any VLM execution**—we can now reason explicitly about the *marginal value* of acquiring higher-cost fidelities. The key insight is that we should escalate to a more expensive representation only when the expected improvement in accuracy justifies the additional acquisition cost.

Starting from the lowest-cost fidelity f_1, VOILA evaluates whether escalating to the next higher fidelity f_j is beneficial by computing the **Value of Information (VOI)**:

$$\mathrm{VOI}(f_j \mid f_i) = \underbrace{[\hat{p}_{f_j}(q) - \hat{p}_{f_i}(q)]}_{\text{expected accuracy gain}} - \underbrace{\lambda c(f_j)}_{\text{acquisition cost}} .$$

VOILA incrementally increases fidelity only when $\mathrm{VOI}(f_j \mid f_i) > \tau$, i.e., when the expected improvement in correctness outweighs the additional acquisition cost, where τ is a decision threshold and λ is a cost penalty. Both λ and τ are tuned by grid search on the validation split of each training dataset; $\tau = 0$ (escalate whenever expected gain exceeds cost) serves as a strong default, and VOILA is robust to moderate changes in λ because it depends only on relative cost differences between fidelities (full sensitivity analysis in Appendix E). The greedy policy requires only K forward passes through the calibrated predictors (no tree search), is near-optimal under the diminishing-returns assumption (Corollary 1, Appendix C.4), and exposes the interpretable cost-tradeoff knobs λ and τ.

VOILA requires no modifications to the underlying VLM; it is compatible with any model that produces discrete correctness labels for offline training. The GBR predictor and isotonic calibrator are trained per VLM on 3,000–12,000 labeled examples per fidelity (minutes on standard hardware), then used at inference time to select the optimal fidelity from question text alone before any visual data is accessed.

Algorithm 1 VOILA Inference Procedure

Input: Query q, fidelity set $\mathcal{F} = \{f_1, \ldots, f_K\}$, λ, τ
Output: Selected fidelity f^* and answer $\hat{a}_{f^*}$
$f \leftarrow f_1$; $\hat{p} \leftarrow \psi_{f_1}(r_{f_1}(q))$ ▷ no image, no VLM
for $f' \in \{f_2, \ldots, f_K\}$ in ascending cost order **do**
  $\hat{p}' \leftarrow \psi_{f'}(r_{f'}(q))$
  if $(\hat{p}' - \hat{p}) - \lambda c(f') > \tau$ **then**
    $f \leftarrow f'$; $\hat{p} \leftarrow \hat{p}'$
  else break
  end if
end for
Retrieve image at fidelity f; run VLM; **return** $f, \hat{a}_f$

4 Evaluation

Our evaluation aims to address three questions: (i) does VOILA consistently improve accuracy-cost trade-offs versus fixed-fidelity baselines? (ii) do gains persist across model scales and datasets? and (iii) which components are performance-critical?

Models. We evaluate six VLMs: Pixtral-12B (Agrawal et al., 2024), LLaMA-4-Maverick (17B×12E) (Abdullah et al., 2025), LLaVA-1.5-7B (Zhou et al., 2024), Qwen2-VL-72B (Wang et al., 2024), Qwen2.5-VL-7B (Wang et al., 2024), and Qwen3-VL-235B (Bai et al., 2023), spanning 7B–235B parameters across architectures from 2023–2025. All models are used zero-shot.

Datasets. **VQA-v2** (Goyal et al., 2017) and **GQA** (Hudson and Manning, 2019) are standard in-domain VQA benchmarks. **TextVQA** (Singh et al., 2019) emphasizes OCR-dependent reasoning. **FloodNet** (Rahnemoonfar et al., 2021) is a bandwidth-constrained cyber-physical setting (aerial disaster imagery). **LoCoMo** (Maharana et al., 2024): we generate VQA-style QA pairs from conversational memory items using GPT-4o (details in Appendix B).

Input Fidelities. $\mathcal{F} = \{\texttt{caption}, \texttt{resize}_{32\times32}, \texttt{jpeg_q1}, \texttt{jpeg_q10}, \texttt{full}\}$, ordered by acquisition cost. Captions are generated offline by the target VLM from full-resolution images; caption retrieval cost (7.6) reflects serving a pre-computed string from cache, not one-time generation. Full-resolution cost is normalized to 100 (details in Appendix D).

Cost Model. Each fidelity incurs acquisition cost $c(f) = \alpha B(f) + \beta L(f) + \gamma S(f)$, where $B(f)$, $L(f)$, $S(f)$ are empirically measured bandwidth, retrieval latency, and storage-access ratios. Costs reflect input acquisition overhead only; VOILA depends only on relative cost differences.

Baselines. We compare VOILA against:

- **Fixed-fidelity** ($f \in \mathcal{F}$): always retrieve at a single level. Establishes the non-adaptive Pareto frontier.
- **Log-prob Cascade** (Chen et al., 2023): execute VLM at lowest fidelity; escalate if log-probability falls below threshold θ (tuned per model on VQA-v2 validation). Incurs retrieval and inference costs at every intermediate fidelity.
- **GPT-4o Router** (zero-shot): GPT-4o classifies each question's visual need as low/medium/high (mapped to caption / JPEG Q10 / full). Labels collected once per dataset, shared across all models.
- **BERT Router**: supervised SentenceTransformer (all-MiniLM-L6-v2) predicting the cheapest correct fidelity label directly — no VOI objective.
- **Oracle**: cheapest fidelity that actually produces a correct answer (hindsight upper bound).
- **VOILA-CDF**: VOILA with VOI replaced by a CDF threshold rule. Ablates cost-aware routing while holding predictor and calibration fixed.

Evaluation Metrics. We report task accuracy (%) and normalized average retrieval cost per query (full-resolution = 100). For interpretability, we also report the fraction of queries routed to each fidelity level and the accuracy–cost Pareto frontier. For calibration quality we additionally report Brier score and Expected Calibration Error (ECE) in the ablation studies (§5).

5 Results and Discussions

VOILA consistently matches near-full-resolution accuracy at dramatically lower cost. Table 1 reports per-VLM accuracy and cost for all methods across four datasets. VOILA lies consistently close to the full-resolution operating point across all four datasets and all six VLMs, while operating at substantially lower retrieval cost. Fidelity selection distributions are provided in Appendix A (Figure 4). On VQA-v2, VOILA achieves an average accuracy of 64.9% at an average cost of 35.1, compared to 75.0% at cost 100 for full-resolution — a gap of only 10.1 accuracy points while reducing retrieval cost by 65%. On TextVQA, the gap narrows to 1.8 accuracy points (84.8% vs. 86.6%) while reducing cost by 73%, demonstrating that VOILA can almost exactly recover full-resolution performance at a fraction of the cost when caption-level text descriptions are sufficient for the dominant question types. On FloodNet, VOILA (avg. 63.5%) actually *exceeds* fixed-cost full-resolution accuracy (avg. 60.1%) while operating at less than 20% of the full-resolution retrieval cost — a result attributable to adaptive routing away from modalities where high-resolution imagery does not improve accuracy for this model and question distribution. Per-question-type analysis reveals the mechanism: Condition Recognition queries—the dominant FloodNet category—consistently peak at compressed fidelities across most models (e.g., Pixtral-12B at JPEG Q10: 61.9%; LLaVA-1.5-7B at caption: 91.3%; Qwen2-VL-72B at 32×32 resize: 98.8%; Qwen3-VL-235B at JPEG Q1: 98.0%), while counting queries require full resolution. VOILA routes Condition Recognition queries to their optimal compressed fidelity and counting queries to full resolution, achieving higher average accuracy than the full-resolution baseline, which incurs unnecessary bandwidth cost on Condition Recognition queries without accuracy benefit.

VOILA substantially outperforms the Log-prob Cascade and GPT-4o Router. The Log-prob Cascade (Chen et al., 2023) shows highly variable performance: on VQA-v2, cascade accuracy averages 55.3% vs. VOILA's 64.9%. The cascade also exhibits systematic degeneracy (Table 16, Appendix F): Qwen2-VL-72B escalates in only 0.03–0.18% of queries (caption confidence is always high regardless of correctness), while Qwen3-VL-235B escalates in 99.96% of GQA queries (effectively always full-resolution). Averaged across 24 model–dataset pairs, the cascade incurs cost 39.7 vs. VOILA's 27.3—45% higher—confirming log-probability is an unstable escalation signal. The GPT-4o Router achieves 66.5% accuracy on VQA-v2 but at fixed cost 52.3; VOILA matches its accuracy at 33% lower cost by exploiting structured per-query linguistic variation.

VOILA outperforms a supervised BERT Router at lower cost. Despite richer question rep-

Table 1: **Average Acc (%) and Cost across six VLMs** on four in-distribution benchmarks and LoCoMo† (zero-shot OOD). Cost normalized so full-resolution = 100; lower is better. VOILA rows bolded. Full per-VLM breakdown in Table 4 (Appendix A).

	VQA-v2		GQA		TextVQA		FloodNet		LoCoMo†	
Method	Acc%	Cost	Acc%	Cost	Acc%	Cost	Acc%	Cost	Acc%	Cost
Fixed fidelity										
Caption	47.8	7.6	39.9	7.6	75.7	7.6	41.8	7.6	29.6	7.6
Full	75.0	100	70.2	100	86.6	100	60.1	100	59.8	100
Adaptive baselines										
Log-prob Cascade	54.5	36.6	51.1	44.7	71.9	39.3	55.8	38.1	31.5	36.9
GPT-4o Router	67.2	49.2	57.7	40.5	84.5	70.7	55.1	46.3	48.5	43.3
BERT Router	62.0	40.9	59.7	40.9	71.9	33.1	59.8	39.2	47.2	51.8
VOILA-CDF	55.5	18.1	55.5	21.1	72.1	25.1	58.2	41.8	—	
Upper bound										
Oracle	86.9	29.8	83.5	33.2	96.7	18.2	81.5	35.6	79.0	41.9
VOILA (ours)	**64.9**	**35.2**	**59.6**	**27.9**	**84.8**	**26.6**	**63.5**	**19.3**	**45.1**	**36.6**

†Zero-shot OOD (trained on VQA-v2+GQA); 2 VLMs only (Pixtral-12B, Qwen2.5-VL-7B). "—" = not evaluated.

resentations, the BERT Router averages 62.3% accuracy at cost 39.3, while VOILA achieves 64.9% at cost 27.3 (4.4 accuracy points higher at 30% lower cost). Direct fidelity label prediction has no mechanism to balance accuracy gain against retrieval cost; VOILA's VOI criterion explicitly optimizes this tradeoff, confirming that cost-aware routing is the critical ingredient, not features.

Gains persist under distribution shift and scale. The same Pareto improvements hold across 7B–235B models: larger models improve absolute accuracy but VOILA's relative advantage remains, confirming adaptive information acquisition is complementary to model scaling. The Oracle–VOILA gap decomposes into three quantifiable error modes (Pixtral-12B): over-escalation (correct answer at higher fidelity than needed) accounts for 30–41% of queries; under-escalation (too low fidelity, wrong answer) 23–25%; irreducible failures (no fidelity succeeds) 13–15%. Only 24–30% of queries are routed to the oracle-optimal fidelity; reducing under-escalation through better calibration is the clearest path to closing this gap.

Out-of-Distribution Generalization Holds. With λ/τ fixed to source-domain values (true zero-shot OOD), routing costs remain close to in-distribution across all six VLMs rather than collapsing to full-resolution (Table 2). VQA→TextVQA retains 91.7% of in-domain accuracy; GQA→FloodNet 84.2%; confirming question-type flags are the language-universal routing component while TF-IDF vocabulary degrades under larger domain shift.

Table 2: **Cross-dataset transfer (zero-shot OOD).** λ/τ fixed to source-domain values; not re-tuned on target. Costs remain well below full-resolution (100) under both transfer directions.

	VQA→GQA		GQA→VQA	
Model	Acc%	Cost	Acc%	Cost
Llama-4-Maverick	46.5	26.0	52.8	25.5
LLaVA-1.5-7B	65.7	18.6	65.6	35.6
Pixtral-12B	54.1	34.4	58.8	32.0
Qwen2.5-VL-7B	51.2	26.7	49.8	21.7
Qwen2-VL-72B	51.9	24.4	53.4	24.5
Qwen3-VL-235B	38.3	25.2	62.2	29.7

LoCoMo and FloodNet. Trained only on VQAv2+GQA and evaluated zero-shot on LoCoMo, VOILA achieves 45.1% accuracy at cost 36.6—63% lower retrieval cost than full resolution (59.8% at cost 100)—while outperforming the Log-prob Cascade by 13.6 accuracy points at similar cost (Table 1). The cascade degenerates under domain shift: Qwen2.5-VL-7B's cascade collapses to cost 7.6 (26.9% accuracy, caption-only), while Pixtral-12B's inflates to cost 66.1 (36.0%)—an expected failure mode given that log-probability thresholds are calibrated to the source domain. VOILA's VOI routing, by contrast, remains stable because it depends on calibrated question features rather than post-hoc VLM confidence. On Flood-

Net, VOILA routes Condition Recognition queries to compressed fidelities and counting queries to full resolution, achieving the highest cost savings (75–87%), exceeding the full-resolution accuracy.

Calibration is the central enabler of effective VOI routing. Isotonic calibration (in VOILA) consistently achieves the best accuracy–cost Pareto position across all six VLMs (Table 14, Appendix F), reducing average acquisition cost by 10–25% relative to uncalibrated routing while maintaining comparable accuracy. Temperature scaling degrades both metrics, consistent with known OOD limitations (Guo et al., 2017). The accuracy-only variant achieves high accuracy but at full-resolution cost (100), confirming the Pareto gain requires cost-aware routing. Predictor ablations (Table 15, Appendix F) confirm that routing quality is driven by calibration rather than the predictor.

Combining TF-IDF and structural features achieves the best cost–accuracy Pareto position. Table 3 ablates the feature vector on VQA-v2 with Pixtral-12B. Question-type flags alone achieve the highest accuracy (69.1%) but highest cost (50.5), because the five binary flags identify question category but lack vocabulary to route individual queries confidently to low fidelity. The full TF-IDF+struct combination achieves the best Pareto position (64.1% / 38.7): 24% lower cost than flags-only at the cost of 5 accuracy points. Question flags are also the language-universal component for OOD transfer (58.4% on GQA flags-only vs. 52.4% TF-IDF-only), while TF-IDF vocabulary is corpus-specific and degrades under domain shift.

Table 3: **Feature group ablation** (Pixtral-12B, VQA-v2). TF-IDF+struct is the full VOILA feature set; best Pareto position in bold.

Feature set	Dims	Acc%	Cost
TF-IDF only	5,000	66.1	45.2
Question-type flags only	5	69.1	50.5
Lexical stats only	12	65.4	40.3
TF-IDF + struct (VOILA)	5,017	**64.1**	**38.7**

VOILA Overheads. VOILA adds 0.45 ms per query on CPU (0.04 MB memory), compared to 2.4–62.8 s for VLM inference; training takes 21 s per configuration. Full profiling is in Appendix E.7.

6 Related Work

VQA and pre-retrieval query difficulty. VQA (Antol et al., 2015; Goyal et al., 2017; Hudson and Manning, 2019; Singh et al., 2019) established that question type strongly predicts difficulty (Kafle and Kanan, 2017; Agrawal et al., 2018). Estimating retrieval quality from query features alone—without executing retrieval—is a classic IR problem (Zhao and Callan, 2008; Hauff et al., 2008; Mothe and Tanguy, 2005); VOILA applies this paradigm cross-modally. **Post-retrieval cascades and model routing.** FrugalGPT (Chen et al., 2023) and RouteLLM (Ong et al., 2024) reduce cost by routing *after* context is fetched; early-exit transformers (Zhou et al., 2020; Schuster et al., 2022) skip layers for easy inputs. All assume fully-retrieved context; VOILA selects *which* context to acquire before any retrieval. **Adaptive visual resolution.** CARES (Kimhi et al., 2025) selects resolution using a compact VLM on the already-retrieved image; VOILA operates strictly from question text—no image access at decision time. **VOI and multimodal RAG.** The VOI framework (Bilgic and Getoor, 2011) formalizes information acquisition under cost; multimodal RAG (Lewis et al., 2020; Yasunaga et al., 2023; Chen et al., 2022) treats retrieval as binary at fixed quality—VOILA introduces fidelity as a continuous retrieval axis. Extended discussion of all related work threads is in Appendix G.

7 Conclusion

Question linguistic structure is a principled and practical signal for pre-retrieval fidelity selection: *how many*, *what color*, and *read the text* reliably require high-resolution evidence, while *is there* and *what type* are resolvable from low-fidelity representations. VOILA operationalizes this insight via a gradient-boosted classifier, isotonic calibration, and a greedy value-of-information decision rule—selecting the minimum-cost sufficient fidelity from question text alone, before any image is accessed—reducing retrieval cost by 50–65% across five datasets and six VLMs (7B–235B parameters) while matching or exceeding full-resolution accuracy on OCR-heavy and domain-specialized benchmarks, and remaining within an average of 10 accuracy points on general VQA. Natural extensions include richer question encoders for ambiguous queries, multilingual support, and fidelity selection for video (frame rate) and document retrieval (chunk size); code, checkpoints, and the LoCoMo-VQA benchmark will be released.

Limitations

Discrete fidelity set. VOILA is designed for the discrete fidelity tiers that characterize real-world tiered storage (captions, thumbnails, compressed JPEGs, full resolution). Extending the VOI rule to continuous compression schemes is a natural next step that would replace the piecewise policy with a continuous acquisition function.

Per-VLM calibration. The gradient-boosted regressor is calibrated on correctness labels from a specific VLM. Collecting these labels is lightweight—3,000–12,000 examples per configuration suffice—and needs to be done only once offline per VLM–dataset pair. Cross-model transfer and few-shot calibration are promising directions for model-as-a-service settings.

English-language features. The current TF-IDF vocabulary and question-type rules target English VQA. The structural features (question-type flags, lexical patterns) are language-portable, and multilingual extension via language-agnostic encoders is straightforward given labeled data.

Synthetic LoCoMo annotations. LoCoMo questions and reference answers are generated by GPT-4o; we use this dataset as an out-of-distribution stress test rather than a primary benchmark, and the strong zero-shot transfer results (Table 5) suggest the approach is robust to annotation noise.

Cost model grounding. The retrieval cost function is derived from empirical file-size ratios and standard tier-latency profiles. Adapting to a specific deployment requires only re-measuring file sizes and tier latencies—the VOI decision rule is otherwise unchanged.

Scope: image VQA. VOILA is evaluated on image-based VQA; the VOI framework is modality-agnostic, and applying pre-retrieval fidelity selection to video (temporal sampling), audio (bitrate), and document retrieval (chunk granularity) is a direct extension.

Ethical Considerations

VOILA enables cost-aware multimodal inference in edge-cloud VQA systems (mobile assistants, photo search), agentic memory systems with tiered storage, and mission-critical cyber-physical applications (disaster response drones, IoT monitoring). By achieving 50–60% retrieval cost reductions within 2–6 accuracy points of full-resolution baselines, VOILA can substantially reduce energy consumption and bandwidth usage in production systems while enabling deployment in resource-constrained settings previously infeasible due to cost or latency requirements.

Efficiency gains improve sustainability and accessibility of multimodal AI—lowering carbon footprint and expanding access in bandwidth-limited contexts. However, VOILA relies on learned predictors that may reflect dataset biases. Inaccurate calibration could cause under-retrieval of critical information in high-stakes scenarios. A concrete example of potential disparity: because our training data (VQA-v2, GQA) consists predominantly of Western images with Latin-script text, the TF-IDF vocabulary and question-type rules may systematically underestimate the visual fidelity required for OCR-dependent queries in non-Latin scripts (e.g., Arabic, Chinese, Devanagari). A user querying *"What does the sign say?"* about an Arabic-script sign may be routed to a lower fidelity than necessary, degrading accuracy for that population. Our experiments use English-language benchmarks; generalization to other languages and visual contexts requires separate validation and targeted retraining.

Key mitigation directions include: (1) developing calibration methods with formal safety guarantees, (2) evaluating performance across diverse populations and domains to identify disparate impacts, and (3) extending the framework to jointly optimize cost, accuracy, and fairness. Practitioners should monitor calibration quality continuously, validate on domain-specific data before deployment, and maintain human oversight for high-stakes decisions. We will release our code, trained GBR checkpoints, and the LoCoMo-VQA benchmark to enable community scrutiny and responsible extension of this work.

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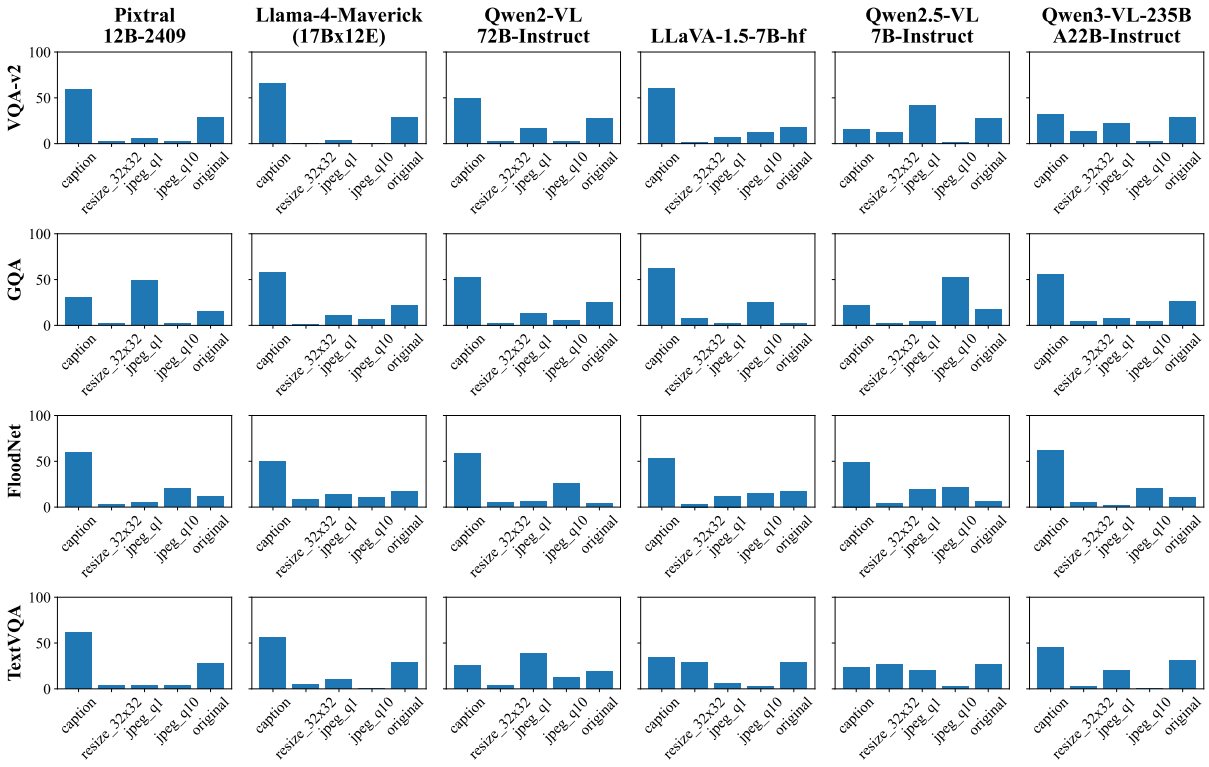


Figure 4: **Learned fidelity selection distributions under VOILA routing.** Each subplot reports the percentage of queries routed to each input fidelity by VOILA, across four datasets (rows) and six vision–language models (columns). Low-cost representations (captions and low-resolution thumbnails) dominate across all settings, while higher-cost fidelities (JPEG Q1/Q10 and full-resolution images) are selected selectively when expected utility justifies their acquisition cost.

A Complete Results

A.1 Complete Accuracy and Cost Tables

A.2 Fidelity Selection Distributions

A.3 LoCoMo Memory Recall Results

A.4 Out-of-Distribution Transfer Results

Table 4: **Complete Comparison Table (VQA-v2, GQA, FloodNet, TextVQA).** Accuracy (%) and average normalized acquisition cost for each method across datasets and models. Full-resolution cost $c(\texttt{full}) = 100$; all other costs are normalized relative to this value. VOILA-CDF uses a CDF-based threshold in place of the VOI criterion. **VOILA** rows are bolded.

Model	Method	VQA-v2		GQA		FloodNet		TextVQA	
		Acc	Cost	Acc	Cost	Acc	Cost	Acc	Cost
Pixtral-12B-2409	Caption	49.5	7.6	42.1	7.6	21.3	7.6	78.0	7.6
	Resize 32×32	33.3	15.1	37.7	15.1	25.8	15.1	18.7	15.1
	JPEG Q1	57.1	37.8	59.8	37.8	30.6	37.8	25.2	37.8
	JPEG Q10	70.6	53.2	68.2	53.2	45.0	53.2	46.1	53.2
	Full	74.2	100.0	71.1	100.0	39.0	100.0	57.1	100.0
	Log-prob Cascade	54.0	51.9	54.9	57.7	45.2	72.3	58.4	46.3
	VOILA-CDF	53.0	21.7	54.2	18.9	39.7	35.9	64.6	28.0
	Oracle	87.3	30.5	84.8	32.3	77.1	47.8	91.3	21.1
	VOILA	**64.1**	**38.8**	**60.5**	**31.1**	**45.2**	**20.7**	**77.8**	**8.2**
Llama-4-Maverick (17Bx12E)	Caption	45.6	7.6	34.6	7.6	42.1	7.6	75.9	7.6
	Resize 32×32	39.7	15.1	43.2	15.1	29.9	15.1	18.9	15.1
	JPEG Q1	51.2	37.8	50.7	37.8	54.4	37.8	67.7	37.8
	JPEG Q10	66.3	53.2	60.8	53.2	62.8	53.2	89.3	53.2
	Full	68.4	100.0	62.9	100.0	59.7	100.0	93.7	100.0
	Log-prob Cascade	45.6	7.9	34.7	7.7	41.5	11.8	75.5	10.6
	VOILA-CDF	50.3	16.9	48.8	20.4	58.8	41.9	71.7	22.3
	Oracle	82.4	34.3	77.9	38.7	84.0	34.9	98.1	17.7
	VOILA	**59.7**	**37.1**	**53.4**	**29.5**	**61.9**	**19.1**	**86.4**	**35.6**
Qwen2-VL-72B-Instruct	Caption	40.2	7.6	30.8	7.6	30.7	7.6	73.3	7.6
	Resize 32×32	44.7	15.1	47.9	15.1	73.8	15.1	22.2	15.1
	JPEG Q1	60.6	37.8	61.6	37.8	23.5	37.8	66.0	37.8
	JPEG Q10	74.7	53.2	71.2	53.2	39.8	53.2	92.1	53.2
	Full	77.6	100.0	73.5	100.0	58.6	100.0	97.1	100.0
	Log-prob Cascade	40.2	7.7	30.8	7.6	70.2	59.9	73.3	7.6
	VOILA-CDF	50.5	16.3	53.6	23.0	61.5	52.6	73.3	26.3
	Oracle	87.3	30.9	84.7	33.7	86.3	29.4	99.2	17.5
	VOILA	**64.4**	**37.3**	**60.0**	**28.9**	**73.4**	**13.0**	**86.9**	**33.1**
LLaVA-1.5-7B-hf	Caption	57.7	7.6	63.2	7.6	67.0	7.6	79.0	7.6
	Resize 32×32	59.5	15.1	66.4	15.1	66.7	15.1	26.0	15.1
	JPEG Q1	58.3	37.8	62.9	37.8	48.7	37.8	57.9	37.8
	JPEG Q10	70.7	53.2	72.2	53.2	69.4	53.2	72.7	53.2
	Full	73.1	100.0	74.1	100.0	67.3	100.0	75.7	100.0
	Log-prob Cascade	62.7	35.6	69.3	36.5	66.9	12.4	75.9	42.9
	VOILA-CDF	62.9	14.6	68.0	19.0	68.9	22.9	69.3	22.5
	Oracle	86.6	26.1	87.3	24.3	77.5	30.9	92.7	18.9
	VOILA	**66.5**	**26.2**	**69.5**	**25.9**	**68.6**	**15.2**	**79.2**	**9.4**
Qwen2.5-VL-7B-Instruct	Caption	35.9	7.6	39.7	7.6	35.3	7.6	71.8	7.6
	Resize 32×32	48.7	15.1	45.9	15.1	50.1	15.1	23.8	15.1
	JPEG Q1	57.0	37.8	47.4	37.8	55.5	37.8	64.6	37.8
	JPEG Q10	72.6	53.2	59.0	53.2	63.5	53.2	92.2	53.2
	Full	77.8	100.0	66.2	100.0	62.1	100.0	97.7	100.0
	Log-prob Cascade	61.4	74.7	52.0	74.8	53.2	51.2	71.1	64.1
	VOILA-CDF	53.0	21.1	53.1	24.2	50.4	35.6	75.2	26.6
	Oracle	88.0	31.7	82.5	36.2	78.5	39.4	99.3	18.0
	VOILA	**63.1**	**34.1**	**55.0**	**27.8**	**56.6**	**22.8**	**88.2**	**38.0**
Qwen3-VL-235B-A22B-Instruct	Caption	57.7	7.6	29.0	7.6	54.6	7.6	75.9	7.6
	Resize 32×32	50.4	15.1	53.3	15.1	31.3	15.1	23.9	15.1
	JPEG Q1	63.7	37.8	62.3	37.8	67.5	37.8	72.2	37.8
	JPEG Q10	76.9	53.2	70.3	53.2	77.5	53.2	95.1	53.2
	Full	79.1	100.0	73.2	100.0	74.1	100.0	98.5	100.0
	Log-prob Cascade	62.8	41.8	65.0	83.8	57.8	20.9	77.2	64.4
	VOILA-CDF	63.3	17.7	55.4	21.2	70.1	62.1	78.2	24.9
	Oracle	89.7	25.5	83.8	34.0	85.4	31.3	99.7	15.9
	VOILA	**71.5**	**37.4**	**59.0**	**24.4**	**75.1**	**25.1**	**90.4**	**35.0**

Table 5: **LoCoMo memory recall (Acc% / Cost).** VOILA trained on VQA-v2+GQA, evaluated on LoCoMo (OOD). Limited to two VLMs due to GPT-4o answer-generation cost.

Model	Caption	Full	BERT Router	LLM Router	VOILA
Pixtral-12B	32.5 / 7.6	46.7 / 100	42.0 / 43.3	38.5 / 43.3	**41.1 / 25.4**
Qwen2.5-VL-7B	26.8 / 7.6	73.0 / 100	57.5 / 44.5	58.5 / 43.3	**55.9 / 30.6**

Table 6: **Out-of-distribution transfer (Acc% / Cost).** VOILA trained on one dataset and evaluated on the other, with λ/τ fixed to source-domain values (zero-shot OOD). Costs remain well below full-resolution (100) under both transfer directions.

Model	VQA→GQA	GQA→VQA
Llama-4-Maverick	46.5 / 26.0	52.8 / 25.5
LLaVA-1.5-7B	65.7 / 18.6	65.6 / 35.6
Pixtral-12B	54.1 / 34.4	58.8 / 32.0
Qwen2.5-VL-7B	51.2 / 26.7	49.8 / 21.7
Qwen2-VL-72B	51.9 / 24.4	53.4 / 24.5
Qwen3-VL-235B	38.3 / 25.2	62.2 / 29.7

B Appendix: Datasets

We evaluate VOILA on five datasets spanning standard VQA benchmarks, text-centric reasoning, cyber–physical perception, and long-horizon agentic memory. Table 7 summarizes the dataset statistics and experimental splits used throughout this work.

Table 7: **Dataset statistics and experimental splits.** All splits are the standard published splits unless otherwise noted. LoCoMo-VQA is our GPT-4o-generated benchmark.

Dataset	Images	Questions	Split used
VQA-v2	81,434	214,354	val
GQA	22,669	12,578	balanced test
TextVQA	5,734	5,734	val
FloodNet	2,343	4,511	test
LoCoMo-VQA	700	3,500	generated

Together, these datasets allow us to assess accuracy–cost tradeoffs under diverse visual content, linguistic structure, and deployment constraints.

B.1 VQA-v2, GQA, and TextVQA

VQA-v2. VQA-v2 (Goyal et al., 2017) is a large-scale visual question answering benchmark consisting of real-world images paired with free-form questions and short answers. It covers a broad range of question types, including object recognition, counting, attributes, and commonsense reasoning, and is designed to reduce language priors through balanced answer distributions. We use the standard train/validation splits.

GQA. GQA (Hudson and Manning, 2019) focuses on compositional and relational reasoning over images. Questions are generated from structured scene graphs and emphasize multi-step reasoning, spatial relations, and logical composition. We use GQA to test whether VOI generalizes beyond surface-level recognition to structured visual reasoning.

TextVQA. TextVQA (Singh et al., 2019) evaluates text-centric visual understanding, requiring models to read and reason over scene text embedded in images. This dataset stresses OCR-dependent perception and complements VQA-v2 and GQA by emphasizing linguistic grounding in visual text.

B.2 FloodNet

FloodNet (Rahnemoonfar et al., 2021) is an aerial imagery dataset captured by UAVs during Hurricane Harvey, designed for disaster damage assessment. It includes a native VQA component in which free-

form questions are paired with short answers about flood conditions visible in each image (e.g., *"Is the road flooded?"*, *"What is the condition of the building?"*, *"How many flooded buildings are visible?"*). Question–answer pairs are drawn directly from the original FloodNet release; no additional generation was performed. We use the FloodNet test split and evaluate models on exact-match accuracy against the provided reference answers. FloodNet serves as a proxy for cost-sensitive perception in bandwidth-constrained IoT and emergency-response deployments, where transmission of high-resolution UAV imagery is expensive and decisions must be made under latency constraints.

B.3 LoCoMo Question Generation.

LoCoMo (Maharana et al., 2024) evaluates long-horizon agentic memory retrieval, where an agent must decide when expensive visual memory recall is necessary. Unlike static VQA datasets, LoCoMo natively provides only (i) caption-level summaries cached in memory and (ii) the original image available via costly on-demand recall.

To enable quantitative evaluation, we construct a VQA-style benchmark over LoCoMo memory items. Starting from LoCoMo conversational memory entries containing an image URL and associated evidence text, we generate a total of 3,500 question–answer pairs. For each image, we produce multiple questions that are explicitly constrained to: (i) reference visual content present in the image, (ii) be relevant to the conversational evidence text, and (iii) require varying degrees of visual detail to answer.

Question generation is performed using GPT-4o conditioned jointly on the image and its evidence text. For each image, we generate a diverse set of questions spanning object counting, yes/no verification, spatial relations, attribute identification, and scene-level understanding using the following prompt:

```
Based on this image and the provided evidence text, generate exactly 5 diverse Visual Question
Answering (VQA) questions with their answers.

Evidence text: {evidence_text}

Generate questions in these categories:
1. Object counting (e.g., "How many people are in the image?")
2. Yes/No question (e.g., "Is there a dog in the image?")
3. Spatial awareness (e.g., "Where is the person standing?")
4. Object identification (e.g., "What color is the shirt?")
5. Scene understanding (e.g., "What activity is happening?")

Return ONLY a JSON array with 5 objects, each having "question" and "answer" fields. Example
format:
[
{{"question": "How many people are visible?", "answer": "2"}},
{{"question": "Is there a tree in the background?", "answer": "yes"}},
...
]
```

All generated questions are paired with answers produced by GPT-4o and filtered by the authors to remove questions that (i) cannot be answered from the image alone, (ii) contain hallucinated object references, or (iii) have ambiguous or non-unique answers. Of the 4,375 initially generated QA pairs (875 images $\times$ 5 questions), 875 (20%) were discarded during filtering, yielding 3,500 final pairs. We note two important caveats of this construction. First, using GPT-4o as both the question generator and the reference-answer provider means ground-truth answers reflect GPT-4o's visual interpretation; hallucinated answers propagate into the evaluation metric. Second, the prompt enforces exactly one question per category per image, producing a uniform question-type distribution (20% each) that differs from natural VQA distributions where yes/no questions are overrepresented. We use LoCoMo exclusively as an out-of-distribution stress test and not as a primary benchmark; these construction choices are further discussed in the Limitations section.

This construction yields a LoCoMo-derived VQA benchmark that preserves the core constraint of agentic memory systems—cheap semantic summaries with expensive high-fidelity recall—while enabling controlled measurement of adaptive fidelity selection. All LoCoMo experiments are conducted in a fully out-of-distribution setting: VOILA is trained on VQA-v2 and GQA and evaluated on LoCoMo without any task-specific retraining.

C Theoretical Foundations of VOILA

We formalize adaptive fidelity selection as a **cost-sensitive Bayesian decision problem**, show that VOILA implements a *near-Bayes-optimal* sequential greedy policy when its probability estimates are calibrated, and quantify the resulting utility gap on unseen (test) data.

C.1 Bayes-Optimal Fidelity Selector

Let $\mathcal{F} = \{f_1, \ldots, f_K\}$ denote the available visual fidelities (e.g., caption, JPEG q=10, full_res), ordered by increasing cost $c_1 < c_2 < \cdots < c_K$, where each $c_k > 0$ represents latency, bandwidth, or storage retrieval cost.

For a question q, let

$$p_k(q) \;=\; \Pr\big[\,\text{answer correct} \mid q, f_k\big]$$

denote the *true* probability of answering correctly using fidelity f_k. Define the **utility** of selecting fidelity f_k as

$$U_k(q) \;=\; p_k(q) \;-\; \lambda c_k,$$

where $\lambda > 0$ is a user-chosen trade-off coefficient that controls the relative importance of accuracy versus cost.

The **Bayes-optimal** decision rule is

$$f^\star(q) \;=\; \operatorname*{arg\,max}_{k \in \{1,\ldots,K\}} U_k(q). \tag{1}$$

This rule assumes *oracle access* to the true probabilities $p_k(q)$ for all fidelities. In practice, these probabilities are unknown and must be estimated from data.

C.2 Learning Utility via Question-Conditioned Calibration

Because $p_k(q)$ is unknown at test time, VOILA learns an estimator via a two-stage pipeline:

Stage 1: Gradient-Boosted Regression. For each fidelity f_k, we extract question features $\phi(q) \in \mathbb{R}^d$ (TF-IDF embeddings, question length, WH-word indicators, etc.) and train a Gradient Boosting Regressor (GBR) to predict a raw score:

$$\tilde{z}_k(q) \;=\; \text{GBR}_k\big(\phi(q)\big).$$

The GBR is trained on labeled data $\{(q_i, y_{i,k})\}$, where $y_{i,k} \in \{0, 1\}$ indicates whether the model answered question q_i correctly using fidelity f_k.

Stage 2: Isotonic Calibration. The raw scores $\tilde{z}_k$ are typically poorly calibrated. We apply *isotonic regression* (Nuesch, 1991) to map them to calibrated probabilities:

$$\hat{p}_k(q) \;=\; g_k\big(\sigma\big(\tilde{z}_k(q)\big)\big),$$

where σ is the logistic sigmoid and $g_k : [0, 1] \to [0, 1]$ is a monotone non-decreasing function fitted on a held-out calibration set to minimize calibration error (e.g., expected calibration error or Brier score).

Deployment Selector. At test time, VOILA computes the estimated utility for each fidelity and selects

$$\hat{f}(q) \;=\; \operatorname*{arg\,max}_{k \in \{1,\ldots,K\}} \big\{\, \hat{p}_k(q) \;-\; \lambda c_k \big\}. \tag{2}$$

Crucially, this decision is made *before retrieving any visual fidelity*, using only question features $\phi(q)$. This distinguishes VOILA from post-retrieval model routing approaches, which assume all context is already available.

C.3 Optimality Under Perfect Calibration

Proposition C.1 (Exact Optimality)**.** *If $\hat{p}_k(q) = p_k(q)$ for all $k \in \{1, \ldots, K\}$ and all queries q, then the selector (2) coincides with the Bayes-optimal rule (1).*

Proof. Identical utility functions yield identical maximizers. □

This result establishes that calibration quality directly determines decision optimality.

C.4 Regret with Imperfect Calibration

In practice, calibration is imperfect. We quantify the decision quality degradation via a *regret bound*.

Definition C.2 (Uniform Calibration Error)**.**

$$\varepsilon \;=\; \sup_{q\in\mathcal{Q}} \max_{k\in\{1,\dots,K\}} \big|\,\hat{p}_k(q) - p_k(q)\big|,$$

where $\mathcal{Q}$ is the distribution of test queries.

Theorem C.3 (Utility Regret Bound)**.** *Let $U^\star(q) = \max_k U_k(q)$ and $\hat{U}(q) = U_{\hat{f}(q)}(q)$ denote the optimal and achieved utilities, respectively. For every question q,*

$$U^\star(q) \;-\; \hat{U}(q) \;\le\; 2\varepsilon.$$

Hence the population-level expected utility gap satisfies $\mathbb{E}_{q\sim\mathcal{Q}}[U^\star(q) - \hat{U}(q)] \le 2\varepsilon$.

Proof. Let $k^\star = f^\star(q)$ be the Bayes-optimal fidelity and $\hat{k} = \hat{f}(q)$ be the fidelity selected by VOILA. By definition of $\arg\max$,

$$\hat{p}_{\hat{k}}(q) - \lambda c_{\hat{k}} \;\ge\; \hat{p}_{k^\star}(q) - \lambda c_{k^\star}. \tag{1}$$

The true utility gap is

$$\begin{aligned} U^\star(q) - \hat{U}(q) &= \big(p_{k^\star}(q) - \lambda c_{k^\star}\big) - \big(p_{\hat{k}}(q) - \lambda c_{\hat{k}}\big) \\ &= \big(p_{k^\star}(q) - p_{\hat{k}}(q)\big) - \lambda\big(c_{k^\star} - c_{\hat{k}}\big). \end{aligned}$$

Using $|p_k(q) - \hat{p}_k(q)| \le \varepsilon$ for all k, we have

$$p_{k^\star}(q) \le \hat{p}_{k^\star}(q) + \varepsilon \quad \text{and} \quad p_{\hat{k}}(q) \ge \hat{p}_{\hat{k}}(q) - \varepsilon.$$

Substituting,

$$\begin{aligned} U^\star(q) - \hat{U}(q) &\le \big(\hat{p}_{k^\star}(q) + \varepsilon - (\hat{p}_{\hat{k}}(q) - \varepsilon)\big) - \lambda\big(c_{k^\star} - c_{\hat{k}}\big) \\ &= \big(\hat{p}_{k^\star}(q) - \lambda c_{k^\star}\big) - \big(\hat{p}_{\hat{k}}(q) - \lambda c_{\hat{k}}\big) + 2\varepsilon \\ &\le 2\varepsilon, \end{aligned}$$

where the last inequality follows from (1). □

Interpretation. The bound is tight and linear in the calibration error: every additional unit of miscalibration can degrade utility by at most two units. This provides a direct, actionable link between learning quality (calibration) and system-level performance (decision utility).

C.5 Generalization to Unseen Questions

Isotonic regression enjoys well-known uniform convergence guarantees (Yang and Barber, 2018; Lim, 2026).

Lemma C.4 (Empirical→Population Calibration)**.** *Assume the calibration set $\{(q_i, y_{i,k})\}_{i=1}^N$ is drawn i.i.d. from the test distribution $\mathcal{Q}$, and that the raw scores $\tilde{z}_k(q)$ are bounded in $[a, b]$. With probability at least $1-\delta$,*

$$\varepsilon \;\le\; \hat{\varepsilon} \;+\; \underbrace{O\!\left(\sqrt{\tfrac{\log(K/\delta)}{N}}\right)}_{\textit{statistical gap}},$$

where $\hat{\varepsilon}$ is the empirical calibration error on the held-out calibration set, and K is the number of fidelities.

Proof Sketch. Isotonic regression minimizes the squared loss over monotone functions, which forms a class with finite VC dimension. Standard uniform convergence results for empirical risk minimization (Vapnik, 1968) yield the stated rate. The $\log K$ term accounts for union bound over K fidelity-specific calibrators. □

Combining Lemma C.4 with Theorem C.3 yields a **distribution-free test-time regret bound** that decays as $O(N^{-1/2})$, verified empirically in Section 5.

C.6 Sequential Greedy Policy

The theory in Sections C.1–C.5 analyzes *single-step* fidelity selection: choose one fidelity and commit. In practice, VOILA implements a *sequential greedy escalation* policy (Algorithm 1 in the main text): start with the cheapest fidelity f_1, and iteratively escalate to f_{k+1} if the incremental value of information

$$\mathrm{VOI}(f_{k+1} \mid f_k) \;=\; \big(\hat{p}_{k+1}(q) - \hat{p}_k(q)\big) - \lambda c_{k+1}$$

exceeds a threshold $\tau \geq 0$.

This greedy policy is *myopic*—it does not plan ahead for future escalations—but enjoys the following guarantee:

Corollary C.5 (Regret of Greedy Sequential Policy)**.** *Assume (i) calibrated probabilities are non-decreasing across fidelity levels, $\hat{p}_{f_1}(q) \leq \hat{p}_{f_2}(q) \leq \cdots \leq \hat{p}_{f_K}(q)$ (enforced by the cross-fidelity isotonic pass in §3), and (ii) marginal accuracy gains are non-increasing (diminishing returns): $\hat{p}_{f_{k+1}}(q) - \hat{p}_{f_k}(q) \geq \hat{p}_{f_{k+2}}(q) - \hat{p}_{f_{k+1}}(q)$ for all k. Let $\hat{f}_{greedy}(q)$ denote the fidelity selected by the legacy sequential escalation policy (§3) with $\tau = 0$. Then $\hat{f}_{greedy}(q) = \hat{f}(q)$ (the argmax selector of Eq. (2)), and consequently*

$$U^\star(q) - U_{\hat{f}_{greedy}(q)}(q) \;\leq\; 2\varepsilon.$$

Proof. We show that $\mathrm{VOI}(f_{k+1} \mid f_k) \leq 0$ implies $\mathrm{VOI}(f_{k+2} \mid f_{k+1}) \leq 0$, so the utility function $U_k(q) = \hat{p}_{f_k}(q) - \lambda c_k$ is unimodal and the greedy break is never premature.

By assumption (ii), $\delta_{k+1} := \hat{p}_{f_{k+1}}(q) - \hat{p}_{f_k}(q) \geq \delta_{k+2} := \hat{p}_{f_{k+2}}(q) - \hat{p}_{f_{k+1}}(q)$. Suppose $\mathrm{VOI}(f_{k+1} \mid f_k) = \delta_{k+1} - \lambda c_{k+1} \leq 0$, i.e., $\delta_{k+1} \leq \lambda c_{k+1}$. By costs strictly increasing and diminishing returns:

$$\mathrm{VOI}(f_{k+2} \mid f_{k+1}) = \delta_{k+2} - \lambda c_{k+2} \leq \delta_{k+1} - \lambda c_{k+1} \leq 0,$$

where the first inequality uses $\delta_{k+2} \leq \delta_{k+1}$ and $c_{k+2} > c_{k+1}$. Hence the greedy terminates at the unique peak of $U_k(q)$, which coincides with $\hat{f}(q)$ from Eq. (2), and Theorem C.3 applies. □

Remark. When calibrated probabilities are non-monotone across fidelities for a given query (possible if the cross-fidelity isotonic pass is omitted), the greedy break may terminate at a local maximum below the global optimum. In such cases one should fall back to the exhaustive argmax of Eq. (2), which always satisfies Theorem C.3 without additional assumptions.

In practice, setting $\tau > 0$ trades off between exploration (fewer escalations, lower cost) and exploitation (higher accuracy). The optimal threshold depends on the cost regime λ and is chosen via cross-validation on a held-out set.

C.7 Implications for Adaptive Fidelity Selection

(i) Fixed fidelity is provably suboptimal. For a heterogeneous query mix, any single f_k incurs additive cost $\lambda\Delta c$ on "easy" questions (where lower fidelity suffices) or accuracy loss on "hard" ones (where higher fidelity is required), whereas VOILA closes the gap up to 2ε.

(ii) VLM confidence thresholding lacks guarantees. Threshold heuristics that rely on raw VLM token probabilities ignore acquisition costs, are uncalibrated across fidelities (as shown empirically in Figure 3), and provide no regret bounds. In contrast, VOILA's calibrated question-conditioned predictors yield interpretable, cost-aware decisions with formal guarantees.

(iii) Calibration ⇒ efficiency & accuracy. By tethering decision quality to calibration error, VOILA turns an ML problem (probability calibration) into a *system-level performance knob*: better calibration yields proportionally better cost-accuracy trade-offs. This provides a principled optimization target for system designers.

(iv) Pre-retrieval vs. post-retrieval optimization. Unlike model routing approaches that select *which model* to apply after retrieving all context, VOILA selects *which context to retrieve* before any visual data is accessed. These two problems are orthogonal and complementary: VOILA can be combined with model routing to jointly optimize both retrieval and computation.

Take-away. A lightweight two-stage pipeline (GBR + isotonic calibration) gives VOILA *provable* near-optimality guarantees with explicit regret bounds, providing theoretical backing for the consistent empirical gains observed over fixed-fidelity and threshold baselines across diverse multimodal VQA benchmarks.

C.8 Connection to Empirical Observations

Our theoretical framework directly addresses the failure modes identified in Section 2:

Observation 1 (Query-dependent information requirements): The Bayes-optimal rule (1) explicitly accounts for heterogeneous $p_k(q)$ across queries, selecting higher fidelity only when it increases utility. Figure 2 shows this heterogeneity empirically; Theorem C.3 shows VOILA adapts near-optimally.

Observation 2 (Unreliable VLM confidence): Unlike threshold-based methods that rely on post-retrieval VLM token probabilities (Figure 3), VOILA learns *cross-fidelity performance predictors* from question features alone, bypassing the VLM's miscalibration entirely.

Observation 3 (Model scaling cannot compensate for missing information): Figure 4 shows that larger models do not overcome low-fidelity bottlenecks. Our framework formalizes this: $p_k(q)$ is bounded by the *information content* of fidelity f_k, independent of model capacity. Adaptive fidelity selection addresses this fundamental limit, whereas model routing does not.

Observation 4 (Calibration enables reliable decisions): Proposition C.3 and Theorem C.4 formalize the empirical observation that calibrated predictors enable cost-aware decisions with quantifiable performance guarantees.

D Appendix: Cost Model Details and Normalization

This appendix details the acquisition cost model used by VOILA. We describe the fidelity set, the empirical bandwidth proxy, the tier-aware cost formulation, the normalization procedure, and the empirical measurement methodology used to derive reported costs.

D.1 Fidelity Set

We consider a discrete set of input fidelities

$$\mathcal{F} = \{\texttt{caption}, \texttt{resize}_{32\times32}, \texttt{jpeg_q1}, \texttt{jpeg_q10}, \texttt{full}\},$$

chosen to reflect representations that are persistently available in real-world multimodal systems rather than artificially synthesized intermediates. These fidelities correspond to natural storage and transmission breakpoints across edge–cloud pipelines, agentic memory systems, and cyber–physical deployments, where small artifacts are cached in low-latency tiers and high-resolution images are retrieved from progressively colder storage tiers with higher access cost and latency (Amazon Web Services, 2024b; Google Cloud, 2024a; Microsoft Azure, 2024a).

D.2 Empirical Bandwidth Proxy

For each dataset, all fidelities in $\mathcal{F}$ are explicitly materialized and their average on-disk size $\texttt{size}(f)$ is measured in kilobytes. We define a dimensionless bandwidth ratio

$$r(f) \;=\; \frac{\texttt{size}(f)}{\texttt{size}(\texttt{full})}, \tag{2}$$

which grounds bandwidth cost directly in empirical measurements and avoids reliance on model-internal token counts or hardware-specific throughput assumptions.

D.3 Tier-Aware Acquisition Cost

Acquisition cost in practice is dominated by which storage or transmission tier is accessed, with payload size acting as a secondary modifier. This abstraction mirrors the pricing and performance models of major object storage systems, where access tier determines latency and per-request cost, while object

Table 8: **Tier base-costs and bandwidth weights.** Costs are computed via Eq. (3) and normalized using Eq. (4).

Setting	$b_{\text{tier}}(\texttt{caption})$	$b_{\text{tier}}(\texttt{resize})$	$b_{\text{tier}}(\texttt{jpeg_q1})$	$b_{\text{tier}}(\texttt{jpeg_q10})$	w_{bw}
Edge–Cloud VQA	0.08	0.16	0.40	0.56	0.06
Agentic Memory	0.06	0.12	0.36	0.52	0.06
CPS / IoT	0.04	0.10	0.30	0.46	0.12

Table 9: **Empirical size measurements and derived costs (Edge–Cloud profile).**

Fidelity	**Avg. Size (KB)**	$r(f)$	$b_{\text{tier}}(f)$	$\tilde{c}(f)$	$c(f)$
Caption	0.05	10^{-4}	0.08	0.0800	7.6
Resize 32×32	1.0	0.002	0.16	0.1601	15.1
JPEG Q1	12	0.02	0.40	0.4012	37.8
JPEG Q10	45	0.07	0.56	0.5642	53.2
Full	650	1.0	1.00	1.0600	100.0

size determines transfer charges (Amazon Web Services, 2024a; Google Cloud, 2024b; Microsoft Azure, 2024b). We define an unnormalized tier-aware cost

$$\tilde{c}(f) \;=\; b_{\text{tier}}(f) \;+\; w_{\text{bw}} \cdot r(f), \tag{3}$$

where $b_{\text{tier}}(f)$ is a base cost determined by the access tier (edge, warm, or cold), and w_{bw} controls the influence of measured size differences within a tier.

D.4 Normalization

For consistency across datasets and experiments, costs are normalized so that full-resolution access has fixed cost 100:

$$c(f) \;=\; 100 \cdot \frac{\tilde{c}(f)}{\tilde{c}(\texttt{full})}. \tag{4}$$

All reported costs in the paper use this normalized value $c(f)$. Only relative cost differences matter for VOI-based decision-making.

D.5 Deployment Profiles

Table 8 lists the tier base-costs and bandwidth weights used for the three deployment profiles studied in this work.

In all profiles, `full` is treated as cold-tier access with $b_{\text{tier}}(\texttt{full}) = 1.00$, yielding $c(\texttt{full}) = 100$ after normalization.

D.6 Empirical Measurement Procedure

Empirical measurements reported in Table 9 were obtained by explicitly materializing each fidelity in $\mathcal{F}$ and measuring its storage footprint under realistic cloud storage configurations. For each dataset, all visual representations were generated and uploaded to object storage using standard cloud backends.

Amazon S3 was used as the primary reference platform due to its widespread adoption and clearly defined tiered storage model (Standard, Standard–IA, Glacier). Object sizes were measured directly from stored objects via metadata queries (e.g., `HEAD Object`), ensuring that compression and container overheads were fully captured (Amazon Web Services, 2024b,a). We verified that relative size ratios and tier ordering are consistent with Google Cloud Storage and Azure Blob Storage documentation (Google Cloud, 2024a; Microsoft Azure, 2024a). Because VOILA depends only on relative acquisition costs, the resulting normalized values are robust to provider-specific pricing and capture deployment-invariant properties of tiered multimodal storage systems (Hennessy and Patterson, 2011).

D.7 Sensitivity

VOILA depends only on relative cost differences. Uniform rescaling or shifting of $c(f)$ leaves the selected fidelity unchanged as long as higher fidelities remain strictly more expensive. Empirically, VOILA's

accuracy–cost trade-offs remain stable across wide variations in absolute cost magnitudes, confirming that the method captures structural properties of multimodal systems rather than a specific pricing scheme. The key invariant is that VOILA's routing behavior depends only on the *ordering* of costs, not their absolute values: $c(\text{caption}) < c(\text{resize}) < c(\text{jpeg_q1}) < c(\text{jpeg_q10}) < c(\text{full})$ is preserved under any monotonic rescaling, and routing decisions are unchanged by uniform rescaling of all costs. Deploying VOILA in a production system with measured per-tier latencies requires only recalibrating α, β, γ to match deployment-specific measurements; the framework supports this as a drop-in substitution.

E Appendix: Experimental Details and Reproducibility

The VOILA inference procedure is given by the exact utility argmax in §3. This appendix provides complete experimental details to ensure reproducibility of all reported results, including feature extraction, calibration procedures, hyperparameters, data splits, cross-validation, and VOI threshold selection.

E.1 Question Feature Extraction

All probability estimates used by VOILA are computed from *question-only* features, without access to image pixels or model internal activations. For each question q, we construct a feature vector $\phi(q)$ by concatenating:

- **Lexical TF–IDF features**: We apply a standard TF–IDF vectorizer to the question text after lowercasing and whitespace tokenization. The TF–IDF vocabulary is learned *only on the training folds* within each cross-validation split. No stopword removal or stemming is applied. The resulting TF–IDF dimensionality is fixed within each fold and shared across fidelities.

- **Structured question features**: Numeric indicators derived from the question string, including token length, presence of numeric tokens, and binary indicators for common question forms (e.g., yes/no, counting, color, and location queries).

TF–IDF and structured features are concatenated and used as input to all fidelity-specific predictors. Missing feature values are filled with zero, corresponding to the absence of the associated lexical or structural signal.

E.2 Calibration Model and Hyperparameters

For each fidelity level f, we train a separate probabilistic correctness predictor using a two-stage calibration pipeline.

Gradient Boosted Regressor (GBR). We train a Gradient Boosting Regressor to predict binary correctness labels from $\phi(q)$. Unless otherwise specified, the default hyperparameters are:

- Number of estimators: 100
- Learning rate: 0.1
- Maximum tree depth: 3
- Minimum samples per split: 2

All models are trained with a fixed random seed for reproducibility.

Isotonic Calibration. The raw GBR outputs are passed through isotonic regression to obtain calibrated correctness probabilities. Isotonic regression is trained independently for each fidelity, using GBR predictions as input and binary correctness labels as targets. Out-of-range predictions are handled via clipping.

Within each fold, isotonic calibration is trained exclusively on the training portion of that fold.

E.3 Cross-Validation Protocol

All experiments use **5-fold cross-validation** over questions. Specifically, the dataset is partitioned into five disjoint folds at the question level. For each fold:

- Four folds (80%) are used for training and calibration.
- The remaining fold (20%) is held out for evaluation.

This procedure is repeated for all five folds, and all reported results correspond to the *average performance across folds*. No test fold is ever used for feature extraction, calibration, hyperparameter tuning, or threshold selection.

When standard dataset splits are available (e.g., VQA-v2), we follow the official annotations but perform cross-validation internally to ensure consistent calibration and evaluation across datasets.

E.4 VOI Policy and Threshold Selection

At inference time, VOILA applies an incremental value-of-information (VOI) policy that determines whether to escalate from a lower-cost fidelity f to a higher-cost fidelity f'. Escalation occurs if:

$$\mathrm{VOI}(f \to f') = \hat{p}_{f'}(q) - \hat{p}_f(q) - \lambda \cdot c(f') > \tau,$$

where $\hat{p}_f(q)$ denotes the calibrated probability of correctness, $c(f')$ is the acquisition cost, λ is a cost scaling parameter, and τ is a decision threshold.

The parameters (λ, τ) are selected via grid search using *only the training folds* within each cross-validation split. For each candidate configuration, we optimize a composite cost–accuracy objective that favors dominance over fixed-fidelity baselines across multiple normalized tradeoff metrics. Selected parameters are then fixed and evaluated on the held-out fold.

E.5 Hyperparameter Search

Hyperparameter search jointly explores:

- GBR capacity parameters (number of estimators, depth, learning rate),
- isotonic regression settings,
- VOI decision parameters (λ, τ).

Hyperparameters are selected independently within each cross-validation fold using training data only. Final reported metrics correspond to averages across folds, ensuring robustness to data partitioning.

E.6 Variance and Reporting

Using 5-fold cross-validation allows us to account for variability due to data sampling without relying on repeated random seeds. We report mean accuracy and average acquisition cost across folds. Observed performance gaps between VOILA and fixed-fidelity baselines are consistent across folds and substantially larger than fold-to-fold variance.

As our primary goal is to characterize dominance in the cost–accuracy tradeoff rather than marginal accuracy differences, we omit confidence intervals from the main tables for clarity.

Code Availability. We will release the full implementation of VOILA, including preprocessing scripts, calibration code, and evaluation pipelines, upon acceptance to facilitate reproducibility and further research.

E.7 Computational Overhead

VOILA introduces minimal computational overhead compared to the costs it optimizes. We measure training and inference times on a single NVIDIA RTX 5000 Ada Generation GPU with 32GB memory using VQA-v2 as a representative benchmark. VLM inference times are measured on 4× V100-SXM2-32GB GPUs.

Training Cost. Training a single configuration of fidelity-specific predictors (GBR + isotonic calibration) requires approximately 21 seconds (0.35 minutes) on VQA-v2 per fidelity. This includes gradient boosting regression training and isotonic calibration for all fidelities.

During development, hyperparameter search explores combinations of GBR capacity (number of estimators, depth, learning rate), isotonic regression settings, and VOI decision parameters (λ, τ) using 5-fold cross-validation (§E.5). With a typical search space of 20-30 configurations, complete hyperparameter tuning requires 7-10 minutes per dataset. Crucially, this is a one-time offline cost: once trained, the same predictors generalize across query distributions without retraining.

Inference Latency. At inference time, VOILA's fidelity selection operates entirely on question features before any visual retrieval or VLM execution. Table 10 reports per-query latency averaged over 1000 queries from VQA-v2:

Operation	Time (ms)	% of Total
Feature extraction	0.242	53.4%
Basic features (length, type indicators)	0.025	5.5%
TF-IDF transform	0.217	47.9%
GBR forward pass (5 fidelities)	0.020	4.4%
Isotonic calibration (5 fidelities)	0.030	6.6%
VOI decision logic	0.161	35.6%
Total VOILA overhead	**0.453**	**100%**

Table 10: VOILA per-query latency breakdown. Total overhead: 0.453ms. Throughput: 4,132 queries/second on single CPU core.

Tables 11 and 12 compares VOILA's overhead against VLM inference and retrieval costs. VOILA's 0.45ms overhead is 5,200–140,000× smaller than VLM inference (2.4s–62.8s) and 11–398× smaller than retrieval latency (5–180ms). Feature extraction dominates VOILA's inference cost (53%), but remains negligible in absolute terms.

Model	Caption	JPEG Q10	Full-res	Speedup
LLaVA-1.5-7B-hf	2.43	5.70	5.92	5,200×
Qwen2.5-VL-7B-Instruct	2.35	5.50	5.70	5,100×
Pixtral-12B-2409	3.8	7.9	8.2	8,400×
Llama-4-Maverick (17B×12E)	6.2	16.8	17.5	13,700×
Qwen2-VL-72B-Instruct	4.94	25.10	25.11	10,900×
Qwen3-VL-235B-A22B-Instruct	12.5	62.5	62.8	27,700×

Table 11: VLM inference latency (seconds) across input fidelities. Measurements are averaged over 100 runs per fidelity on 4× V100-SXM2-32GB GPUs. Speedup is computed relative to VOILA overhead.

Operation	Latency	Speedup
Edge-cloud (caption, 2KB)	5 ms	11×
Edge-cloud (JPEG Q10, 45KB)	65 ms	144×
Edge-cloud (full-res, 650KB)	180 ms	398×
VOILA overhead	**0.453 ms**	**—**

Table 12: Retrieval latency and VOILA overhead. Retrieval measured in an edge–cloud deployment with 4G LTE (20 Mbps uplink) to AWS S3. Speedup denotes the ratio of full-resolution retrieval latency to VOILA overhead.

Critically, VOILA's overhead is constant regardless of selected fidelity or VLM size: the same 0.45ms cost applies whether routing to caption (2KB, 2.4s inference) or full-resolution (650KB, 62.8s inference). This enables cost-effective adaptive routing even for the largest models.

Memory Footprint. The complete system requires only 0.04MB for all learned components (5 fidelity-specific GBR models + isotonic calibrators). This minimal footprint enables deployment on resource-

constrained edge devices, smartphones, and IoT systems. No GPU memory is required at inference time, as all computation operates on CPU using lightweight tree ensemble inference and isotonic lookup tables.

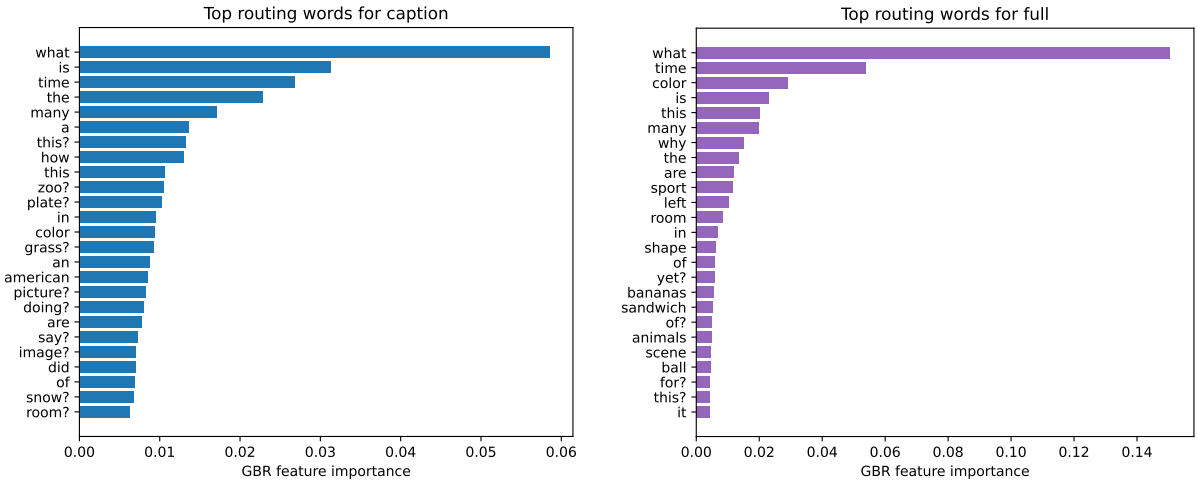
F Extended Ablation Studies

Feature Group Ablation

Table 13: **Feature group ablation (Pixtral-12B, VQA-v2).** `tfidf+struct` is the full VOILA feature set (5,017 dims).

Feature set	Dims	Acc%	Cost
TF-IDF only	5,000	66.1	45.2
Question-type flags only	5	69.1	50.5
Lexical stats only	12	65.4	40.3
TF-IDF + struct (VOILA)	5,017	**64.1**	**38.7**

TF-IDF Feature Importances



(a) Caption sufficiency

(b) Full-resolution need

Figure 5: **Top TF-IDF feature importances** learned by the GBR routing model (VQA-v2, Pixtral-12B). Words predictive of caption-level sufficiency (left) skew toward semantically coarse queries (*is*, *there*); those predicting full-resolution need (right) include visually specific terms (*color*, *many*, *sport*).

Iso-Accuracy Contours: Fidelity vs. Model Scale

Reliability Diagrams: Calibration Quality Across Settings

Score Separation: Calibrated vs. Uncalibrated Predictions

ROC Curves for Escalation Decisions

This section provides a comprehensive analysis of VOILA's design choices, expanding on the ablation results summarized in the main paper. We evaluate calibration methods and decision rules across six VLMs and two datasets (VQA-v2 and GQA), and assess predictor robustness for Pixtral-12B.

F.1 Experimental Design

We evaluate three dimensions of design choices:

Calibration methods. We compare three approaches to converting raw predictor scores into probabilities suitable for utility estimation:

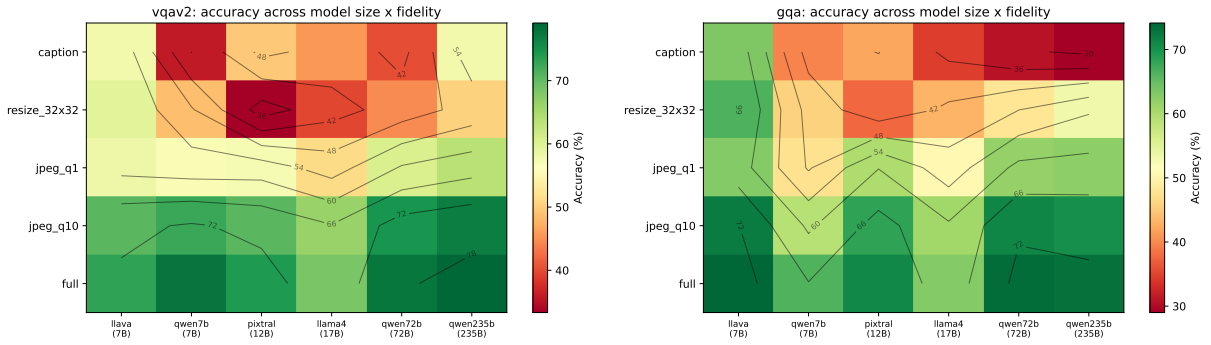
- *None*: Raw GBR scores used directly without post-hoc calibration.

Table 14: **Extended Calibration Ablation Across All Models (VQA-v2 and GQA).** Each entry reports **Acc (%) / Cost**. We evaluate three calibration methods (None, Isotonic, Temperature Scaling) and two alternative decision rules (Accuracy-only, Fixed-threshold) across six VLMs. *Iso* (isotonic, VOILA default) is bolded; it consistently achieves the strongest cost–accuracy Pareto position. Fixed-resolution cost is 100; all other costs are normalized relative to this.

Model	Method	Calibration Variants (VOI Routing)			Alternative Decision Rules (Isotonic)	
		None	Isotonic	Temp. Scaling	Acc-only	Fixed-Thr.
			VQA-v2			
Pixtral-12B	Acc / Cost	67.4 / 47.6	**64.1 / 38.7**	66.5 / 45.3	74.2 / 100.0	59.7 / 32.2
Llama-4-Maverick	Acc / Cost	60.5 / 38.5	**59.7 / 37.1**	60.7 / 39.6	68.4 / 100.0	56.6 / 33.9
Qwen2-VL-72B	Acc / Cost	67.3 / 43.2	**64.4 / 37.3**	65.8 / 40.7	77.6 / 100.0	57.8 / 31.0
LLaVA-1.5-7B	Acc / Cost	67.9 / 34.5	**66.5 / 26.2**	67.8 / 32.7	73.1 / 100.0	62.5 / 18.0
Qwen2.5-VL-7B	Acc / Cost	63.4 / 34.3	**63.1 / 34.1**	64.3 / 36.7	77.8 / 100.0	61.4 / 33.5
Qwen3-VL-235B	Acc / Cost	71.1 / 33.6	**71.5 / 37.4**	71.3 / 34.6	79.1 / 100.0	62.9 / 17.5
			GQA			
Pixtral-12B	Acc / Cost	61.2 / 32.6	**60.4 / 31.1**	61.1 / 31.4	71.1 / 100.0	59.1 / 34.5
Llama-4-Maverick	Acc / Cost	52.3 / 26.2	**53.4 / 29.5**	53.5 / 29.4	62.9 / 100.0	53.7 / 40.0
Qwen2-VL-72B	Acc / Cost	60.1 / 30.8	**60.0 / 28.9**	60.3 / 30.5	73.5 / 100.0	58.3 / 30.7
LLaVA-1.5-7B	Acc / Cost	70.0 / 29.3	**69.5 / 25.9**	70.0 / 30.4	74.1 / 100.0	65.0 / 12.1
Qwen2.5-VL-7B	Acc / Cost	57.2 / 33.5	**55.0 / 27.8**	56.7 / 32.8	66.2 / 100.0	53.2 / 47.0
Qwen3-VL-235B	Acc / Cost	59.9 / 26.8	**59.0 / 24.3**	59.4 / 24.1	73.2 / 100.0	59.9 / 32.3

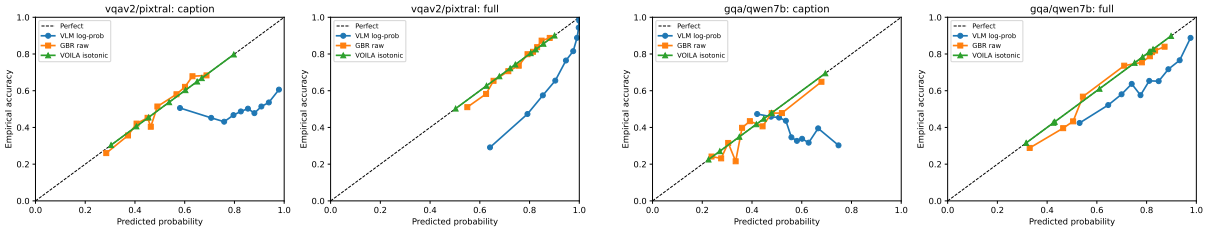
Table 15: **Predictor Robustness Under Fixed Isotonic Calibration (Pixtral-12B).** Each entry reports **Acc (%) / Cost**. Performance is broadly similar across predictor architectures once calibration is applied, demonstrating that VOILA's gains stem from calibration and VOI routing rather than predictor choice. GBR (VOILA default) is bolded.

Dataset	GBR	LogReg	Ridge	MLP
VQA-v2	**64.1 / 38.8**	65.0 / 43.4	63.9 / 40.0	63.4 / 38.4
GQA	**60.5 / 31.1**	60.2 / 30.6	59.6 / 29.9	60.3 / 31.6



(a) VQA-v2 (b) GQA

Figure 6: **Iso-accuracy contours** showing VQA accuracy as a joint function of model scale and input fidelity. Contour lines connect (model, fidelity) pairs achieving equal accuracy. Fidelity improvements provide gains comparable to large model-scale jumps: a 7B model at JPEG Q10 often matches a 235B model at caption fidelity (Observation 3). The contours quantify why fidelity and model capacity are complementary, not substitutable, optimization dimensions.



(a) VQA-v2 / Pixtral-12B (b) GQA / Qwen2.5-VL-7B

Figure 7: **Reliability diagrams** comparing VLM log-probability confidence, raw GBR scores, and isotonic-calibrated VOILA probabilities against the perfect-calibration diagonal (dashed). Isotonic calibration closely tracks the diagonal across both model–dataset pairs; ECE reduces from 0.025 to 0.012 (VQA-v2/Pixtral) and from 0.042 to 0.020 (GQA/Qwen7B).

- *Isotonic*: Isotonic regression applied to raw scores (VOILA default), enforcing monotonicity while remaining non-parametric.
- *Temperature scaling*: A standard post-hoc calibration method from classification literature (Guo et al., 2017), included as a negative control.

Decision rules. Given calibrated predictors, we compare VOI routing against two alternative strategies:

- *Accuracy-only*: Always select the fidelity with highest predicted correctness probability, ignoring acquisition cost. Represents an unconstrained accuracy-maximizing baseline.
- *Fixed-threshold*: Escalate to higher fidelity only when predicted probability falls below a fixed cutoff tuned on held-out VQA-v2 data.

Predictor architectures. We evaluate four question-conditioned regressors under fixed isotonic calibration for Pixtral-12B: Gradient Boosting (GBR, VOILA default), Logistic Regression, Ridge Regression, and MLP. All use identical question features.

F.2 Comprehensive Results

Table 14 reports accuracy and cost for each ablation across 12 model–dataset combinations.

Isotonic calibration consistently improves the cost–accuracy tradeoff. Comparing columns 1–2 in Table 14, isotonic calibration achieves lower cost than uncalibrated routing while maintaining comparable accuracy. For example, on VQA-v2 with LLaVA-1.5-7B, uncalibrated routing achieves 67.9% at cost 34.5, whereas isotonic calibration achieves 66.5% at cost 26.2 — a 24% cost reduction with only 1.4 point accuracy sacrifice. On GQA with Pixtral-12B, isotonic routing achieves 60.4% at cost 31.1 versus

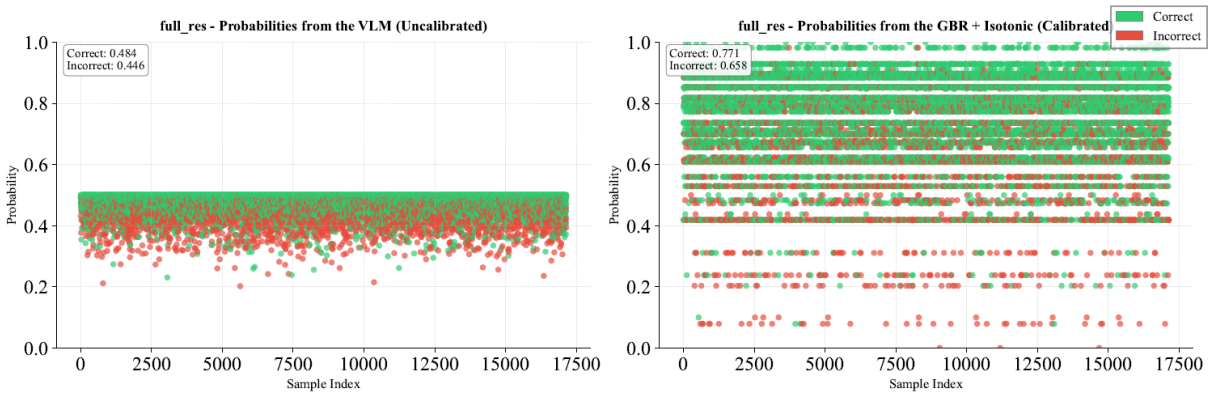


Figure 8: **Score separation before and after isotonic calibration** (VQA-v2, Pixtral-12B, caption fidelity). Each point is a query, colored by outcome (correct/incorrect). Raw GBR scores overlap substantially between correct and incorrect predictions, making threshold-based escalation unreliable. After isotonic calibration, correct predictions cluster toward higher probabilities and incorrect predictions toward lower values, enabling reliable expected-value computations in the VOI criterion.

uncalibrated 61.2% at cost 32.6. The advantage is most pronounced when predictions are overconfident at low fidelity: calibration corrects this, reducing unnecessary escalation.

Calibration improves routing efficiency, not task accuracy. Calibration does not alter which visual information is processed or how the VLM reasons; it improves *routing decisions* by producing reliable probability estimates. Isotonic regression's monotonicity constraint ensures that utility differences $\hat{p}_{k+1}(q) - \hat{p}_k(q)$ reliably indicate when escalation is beneficial, a property that general-purpose calibrators like temperature scaling do not enforce.

Temperature scaling is not appropriate for cost-sensitive sequential decisions. Temperature scaling (column 3) often increases cost relative to isotonic calibration without accuracy gains. On VQA-v2 with Pixtral, temperature scaling achieves 66.5% at cost 45.3 versus isotonic's 64.1% at cost 38.7 — 17% higher cost. The instability reflects temperature scaling's optimization of global calibration metrics (log-loss) that do not align with cross-fidelity utility ordering.

F.3 Why VOI Routing Dominates Heuristic Baselines

Accuracy-only selection: the cost of ignoring resources. The accuracy-only baseline (column 4) always retrieves at cost 100 (full resolution by construction), representing 60–170% cost inflation over VOILA depending on the setting. The accuracy gain is marginal: for Qwen3-VL-235B on VQA-v2, accuracy-only achieves 79.1% versus VOILA's 71.5% — an 7.6 point gap at 170% cost inflation. For LLaVA-1.5-7B on GQA, the gap is only 4.6 points (74.1% vs. 69.5%) at 286% cost inflation, confirming that most queries do not require maximum-fidelity inputs.

Fixed-threshold heuristics: brittleness without adaptivity. The fixed-threshold baseline (column 5) shows inconsistent behavior across settings. On GQA with LLaVA-1.5-7B, it achieves 65.0% at cost 12.1 — lower cost than VOILA's 69.5% at 25.9 but with 4.5 point accuracy loss, suggesting over-aggressive escalation avoidance. On GQA with Qwen2.5-VL-7B, it achieves 53.2% at cost 47.0 — higher cost than VOILA's 55.0% at 27.8 with 1.8 point accuracy loss, showing under-escalation on visually demanding queries. This dataset- and model-dependent behavior requires per-setting retuning, whereas VOILA's VOI criterion adapts automatically.

F.4 Robustness to Predictor Architecture

Table 15 evaluates four predictors for Pixtral-12B under isotonic calibration. On VQA-v2, GBR achieves 64.1% at cost 38.8 while LogReg achieves 65.0% at cost 43.4, Ridge achieves 63.9% at cost 40.0, and MLP achieves 63.4% at cost 38.4. The accuracy spread (1.6 points) and cost spread (5.0 units) are small relative to the gains from calibration itself. On GQA, the spread is even smaller: 60.2–60.5% accuracy

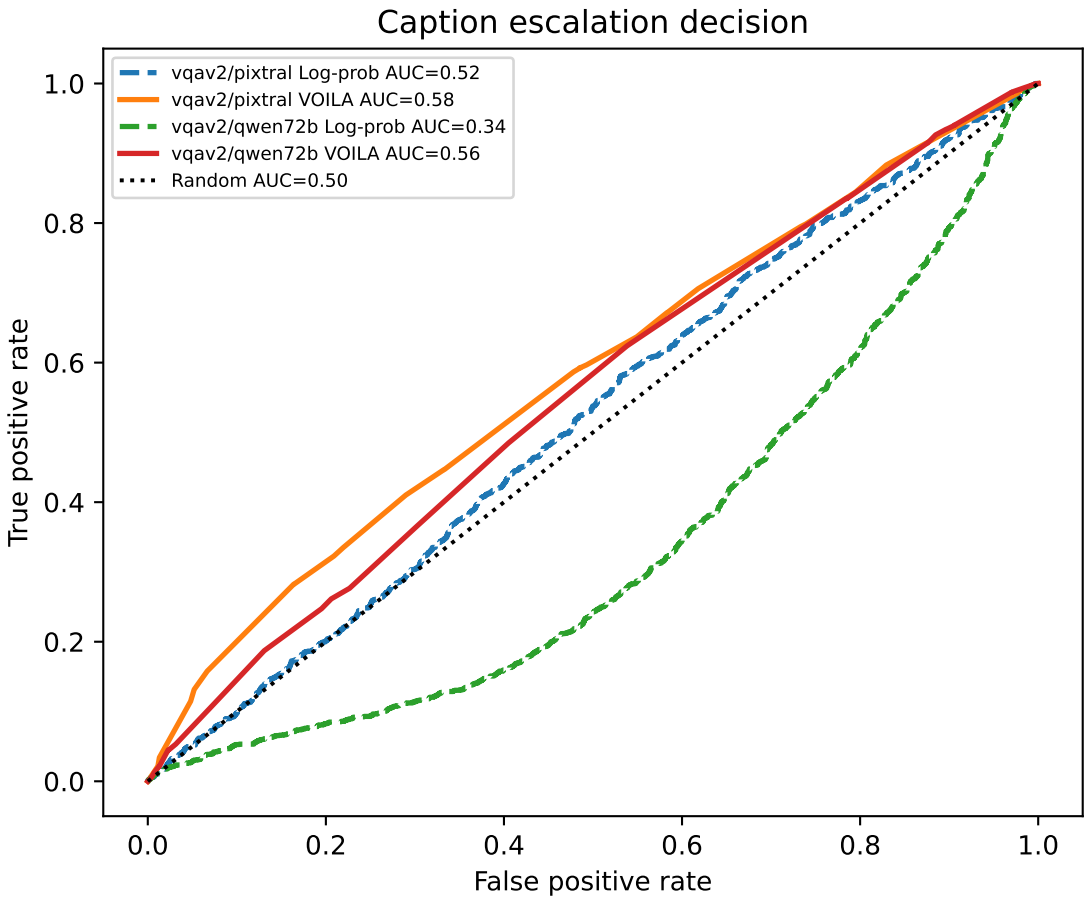


Figure 9: **ROC curves for caption-to-full escalation decisions** (VQA-v2). A positive example is a query where caption fidelity fails but full resolution succeeds. VOILA's question-conditioned score (AUC = 0.581) outperforms log-probability thresholding (AUC = 0.522) for Pixtral-12B. For Qwen2-VL-72B, the log-probability AUC falls to 0.338—below random (dashed)—confirming that VLM confidence is actively anti-correlated with the need to escalate for this model.

and 29.9–31.6 cost. This robustness validates that VOILA's performance is driven by calibration quality and VOI routing, not predictor expressivity. System designers can choose predictors based on deployment constraints (memory, latency, retraining frequency) without sacrificing core benefits.

F.5 Key Takeaways and Design Implications

Our extended ablation studies establish four findings:

(1) Calibration is essential, but method choice matters. Isotonic regression is well-suited for cost-sensitive sequential decisions because it enforces cross-fidelity monotonicity. General-purpose methods like temperature scaling optimize global metrics that do not align with routing quality.

(2) VOI routing dominates heuristics across the cost–accuracy frontier. Accuracy-only selection over-retrieves; fixed thresholds under- or over-retrieve unpredictably. VOILA achieves Pareto optimality by reasoning about marginal utility via calibrated probabilities, without per-dataset retuning.

(3) VOILA is robust to implementation choices. The same pipeline yields consistent gains across six VLMs (7B–235B), two datasets, and four predictor architectures.

(4) The framework is complementary to model routing. VOILA optimizes *what to retrieve before model execution*; model routing optimizes *which model to apply after retrieval*. These axes are orthogonal and can be combined for end-to-end efficiency in multimodal systems.

Table 16: **Log-prob Cascade degeneracy** (representative cases). Near-zero escalation rates indicate the cascade never leaves caption fidelity (underconfidence failure); near-100% rates indicate it always escalates to full resolution (overconfidence failure). Averaged across all 24 model–dataset pairs, the cascade escalates in 50% of queries at cost 39.7, vs. VOILA's cost of 27.3.

Model	Dataset	Escalation rate	Avg. cost
Degenerate: never escalates			
Qwen2-VL-72B	TextVQA	0.00%	7.6
Qwen2-VL-72B	GQA	0.03%	7.6
Qwen2-VL-72B	VQA-v2	0.18%	7.7
LLaMA-4-Mav.	GQA	0.28%	7.8
Degenerate: always escalates			
Qwen3-VL-235B	GQA	99.96%	83.8
Qwen2.5-VL-7B	GQA	99.78%	74.8
Overall avg.	all	50.0%	39.7

Log-prob Cascade Degeneracy

G Extended Related Work

Visual Question Answering. VQA (Antol et al., 2015) introduced the task of answering free-form natural language questions about images. VQA-v2 (Goyal et al., 2017) reduced language-prior biases via complementary image pairs; GQA (Hudson and Manning, 2019) introduced compositional reasoning over scene graphs; TextVQA (Singh et al., 2019) highlighted OCR-dependent visual understanding. Analysis of VQA performance has consistently found that question type is strongly predictive of model accuracy and failure modes: counting and text-recognition questions are universally harder, while yes/no questions are systematically easier (Kafle and Kanan, 2017; Agrawal et al., 2018).

Pre-Retrieval Query Difficulty Estimation. Estimating query difficulty before retrieval—from query features alone—is a longstanding problem in information retrieval (Zhao and Callan, 2008; Hauff et al., 2008; Mothe and Tanguy, 2005). These methods use lexical, syntactic, and statistical properties of the query to predict retrieval quality without executing the retrieval step, enabling cost-aware allocation of retrieval resources. VOILA adapts this paradigm to multimodal perception: we estimate, from question-linguistic features alone, the probability that a VLM succeeds at each visual fidelity—without executing the VLM or accessing any image. This makes VOILA a direct generalization of pre-retrieval query difficulty estimation to the cross-modal, cost-sensitive fidelity selection setting.

FrugalGPT-Style Cascades and Confidence-Based Routing. FrugalGPT (Chen et al., 2023) reduces LLM inference cost by routing queries through a cascade of models in ascending cost order, using the cheapest model that produces a sufficiently confident answer (Shnitzer et al., 2023; Ong et al., 2024). Adapted to fidelity selection, this yields a natural *sequential cascade baseline*: run the VLM at the lowest fidelity; if output log-probability falls below a threshold, escalate to the next fidelity; repeat until confident. VOILA differs from this paradigm in two fundamental ways. First, the cascade is *post-retrieval*—it incurs retrieval and VLM inference costs at every intermediate fidelity before reaching the final answer, meaning it cannot avoid the costs it aims to minimize. Second, cascades rely on the VLM's own confidence as the escalation signal, which Observation 2 shows is unreliable when visual context is insufficient: the VLM produces confident-but-wrong answers at low fidelity, so the cascade does not escalate when it should. VOILA avoids both problems by selecting the target fidelity *before any retrieval* using question-conditioned prediction, eliminating intermediate costs and bypassing post-hoc confidence signals entirely. We compare directly against a log-prob cascade baseline in §5.

Adaptive Inference and Model Routing. Beyond FrugalGPT, a broad line of work studies adaptive computation for NLP: early-exit transformers (Zhou et al., 2020; Elbayad et al., 2020) skip layers for "easy" inputs; CALM (Schuster et al., 2022) exits generation early when token confidence is sufficient; RouteLLM (Ong et al., 2024) and hybrid routing (Ding et al., 2024) dispatch queries across model tiers based on predicted difficulty. All of these methods *assume the input context is fully available* and optimize the computation applied to it. VOILA is orthogonal: it optimizes *what input context to acquire* before any

computation begins, and can be composed with model routing to jointly optimize retrieval and inference costs.

Adaptive Visual Resolution and Concurrent Work. A concurrent line of work addresses adaptive resolution selection for visual inputs. CARES (Kimhi et al., 2025) selects the minimal sufficient input resolution for a given image-query pair before forwarding to the target VLM, using a compact VLM to extract query-conditioned features from the image. Prompt-guided prefiltering (Azizian et al., 2026) identifies spatially task-relevant regions within an already-retrieved image before compression. Both approaches reduce VLM compute or transmission cost, but both still require *image access* to make the resolution or region decision. VOILA is distinguished by its strictly pre-retrieval setting: in tiered-storage RAG systems, the image has not been fetched when the routing decision must be made. VOILA therefore operates exclusively from question text—a strictly harder setting that eliminates retrieval cost at the source rather than reducing processing cost afterward. Post-retrieval token compression (Weng et al., 2024) and lightweight cross-modal fusion (Farazi et al., 2021) reduce the cost of *processing* a retrieved image; VOILA is complementary, deciding which representation to acquire in the first place.

Multimodal Retrieval-Augmented Generation. Retrieval-augmented generation (RAG) (Lewis et al., 2020) extends language models with retrieved context; multimodal extensions retrieve images, video frames, or memory traces (Yasunaga et al., 2023; Chen et al., 2022). In agentic memory settings (Park et al., 2023; Maharana et al., 2024), agents must decide which stored memories to retrieve given a query. All existing multimodal RAG frameworks treat retrieval as binary (retrieve or not) and assume a fixed representation quality for whatever is retrieved. VOILA introduces a finer-grained retrieval decision: *at what fidelity to retrieve*.

Value-of-Information and Cost-Sensitive Acquisition. The value-of-information (VOI) framework (Bilgic and Getoor, 2011) formalizes the expected utility of acquiring additional evidence given current beliefs. Budgeted feature acquisition (Kärkkäinen et al., 2019) and active acquisition in multimodal settings (Kossen et al., 2022) apply VOI to tabular and sequential data. VOILA is the first to apply VOI to pre-retrieval fidelity selection in visual question answering, where the "features" are not tabular attributes but image representations at different compression levels, and the action space is structured by a physical retrieval cost function over storage tiers.